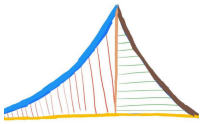
CBSE 2018 Mathematics Class 10 and Class 12 Question Papers

Math for EE

Vishweshwar - ee26btech11225



G. V. V. Sharma

About this Book

This document contains the questions and solutions from the CBSE 2018 Mathematics question papers. Section 1 covers the unique questions from the Class 10 question paper sets. Section 2 covers the Class 12 question paper, Series BVM/5, Code No. 65/5/2 (Set 2).

Questions are typed in their original wording, with the original options and diagrams wherever applicable.

Where the same question appears in multiple sets, it will be included only once, provided the wording, values, options, and diagrams are identical.

CONTENTS

1 CBSE CLASS-10 MATH 2018 4
1.1 Formulae 4
1.2 Problems 11

2 CBSE CLASS-12 MATH 2018 40
2.1 Formulae 40
2.2 Problems 47

1. CBSE CLASS-10 MATH 2018

A. Formulae

1.1.1 If $x=\alpha$ is a root of $p(x)=ax^2+bx+c=0$, then $p(\alpha)=0$, i.e.,

$$\begin{pmatrix}\alpha^2 & \alpha & 1\end{pmatrix}\begin{pmatrix}a\\b\\c\end{pmatrix}=a\alpha^2+b\alpha+c=0 \tag{1.1.1.1}$$

1.1.2 If α and β are distinct zeroes of $p(x)$, then $(x-\alpha)(x-\beta)$ is a factor of $p(x)$.

$$p(x)=(x-\alpha)(x-\beta)q(x) \tag{1.1.2.1}$$

Solution: Dividing $p(x)$ by $(x-\alpha)$,

$$p(x)=(x-\alpha)s(x)+p(\alpha)=(x-\alpha)s(x) \tag{1.1.2.2}$$

since $p(\alpha)=0$. Substituting $x=\beta$,

$$p(\beta)=(\beta-\alpha)s(\beta)=0 \implies s(\beta)=0 \tag{1.1.2.3}$$

as $\beta\neq\alpha$. Repeating the division for $s(x)$ with the zero β gives $s(x)=(x-\beta)q(x)$, yielding (1.1.2.1).

1.1.3 Roots of a quadratic equation $ax^2+bx+c=0$ $(a\neq 0)$ and nature of roots, with discriminant $D=b^2-4ac$:

$$x=\frac{-b\pm\sqrt{D}}{2a} \tag{1.1.3.1}$$

$$\begin{cases} D>0 & \implies \text{two distinct real roots}\\ D=0 & \implies \text{two equal real roots}\\ D<0 & \implies \text{no real roots}\end{cases}$$

Solution: Dividing by a and completing the square,

$$x^2+\frac{b}{a}x+\frac{c}{a}=0 \implies \left(x+\frac{b}{2a}\right)^2=\frac{b^2-4ac}{4a^2} \tag{1.1.3.2}$$

Taking the square root on both sides yields (1.1.3.1). The roots are real only if $D\geq 0$. Further, for roots α,β,

$$\alpha+\beta=-\frac{b}{a},\quad \alpha\beta=\frac{c}{a} \tag{1.1.3.3}$$

which also follows from the fact that α,β are the eigenvalues of the companion matrix

$$\mathbf{C}=\begin{pmatrix}0 & 1\\ -\frac{c}{a} & -\frac{b}{a}\end{pmatrix},\quad \left|\mathbf{C}-\lambda\mathbf{I}\right|=\lambda^2+\frac{b}{a}\lambda+\frac{c}{a} \tag{1.1.3.4}$$

with $\sum\lambda_i=\text{Tr}(\mathbf{C})$ and $\prod\lambda_i=\left|\mathbf{C}\right|$.

1.1.4 The nth term a_n of an Arithmetic Progression (A.P.) with first term a and common

difference d:

$$a_n = a + (n-1)\,d \tag{1.1.4.1}$$

Solution: By definition $a_{k+1} - a_k = d$. Summing for $k = 1, 2, \ldots, n-1$ (telescoping),

$$\sum_{k=1}^{n-1} (a_{k+1} - a_k) = a_n - a_1 = (n-1)\,d \tag{1.1.4.2}$$

and $a_1 = a$, yielding (1.1.4.1).

1.1.5 Sum of the first n terms S_n of an A.P.:

$$S_n = \frac{n}{2}\left[2a + (n-1)\,d\right] = \frac{n}{2}\,(a + l) \tag{1.1.5.1}$$

where $l = a_n$ is the last term.

Solution: Writing the sum forwards and backwards,

$$S_n = a_1 + a_2 + \cdots + a_n \tag{1.1.5.2}$$

$$S_n = a_n + a_{n-1} + \cdots + a_1 \tag{1.1.5.3}$$

From (1.1.4.1), each pair satisfies $a_k + a_{n+1-k} = 2a + (n-1)\,d = a + l$. Adding the two equations,

$$2S_n = n\,(a + l) = n\left[2a + (n-1)\,d\right] \tag{1.1.5.4}$$

yielding (1.1.5.1).

1.1.6 Four consecutive terms in an A.P. with common difference $2d$:

$$\begin{pmatrix} a - 3d \\ a - d \\ a + d \\ a + 3d \end{pmatrix} = a\mathbf{1} + d \begin{pmatrix} -3 \\ -1 \\ 1 \\ 3 \end{pmatrix} \tag{1.1.6.1}$$

where $\mathbf{1}$ is the all-ones vector.

Solution: The difference between any two consecutive entries is

$$(a - d) - (a - 3d) = (a + d) - (a - d) = (a + 3d) - (a + d) = 2d \tag{1.1.6.2}$$

and the mean of the four terms is a, which is why this choice simplifies the equations.

1.1.7 Distance between the points $\mathbf{P} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ and $\mathbf{Q} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$:

$$d = \|\mathbf{Q} - \mathbf{P}\| = \sqrt{(\mathbf{Q} - \mathbf{P})^{\top}(\mathbf{Q} - \mathbf{P})} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \tag{1.1.7.1}$$

Solution: The direction vector from $\mathbf{P}$ to $\mathbf{Q}$ is

$$\mathbf{Q} - \mathbf{P} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix} \tag{1.1.7.2}$$

and its length is the norm, obtained from the Pythagoras theorem, yielding (1.1.7.1).

1.1.8 Section formula: the point $\mathbf{R}$ dividing the line joining $\mathbf{P}$ and $\mathbf{Q}$ internally in the

ratio $k:1$ is

$$\mathbf{R} = \frac{k\mathbf{Q}+\mathbf{P}}{k+1} \iff \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{k+1}\begin{pmatrix} kx_2+x_1 \\ ky_2+y_1 \end{pmatrix} \tag{1.1.8.1}$$

Solution: Since $\mathbf{R}$ lies on the line segment PQ with $PR:RQ = k:1$,

$$\mathbf{R}-\mathbf{P} = k\left(\mathbf{Q}-\mathbf{R}\right) \tag{1.1.8.2}$$

$$\implies (k+1)\,\mathbf{R} = k\mathbf{Q}+\mathbf{P} \tag{1.1.8.3}$$

yielding (1.1.8.1). For $k=1$, $\mathbf{R}$ is the midpoint of PQ.

1.1.9 The area of a triangle with vertices $\mathbf{A} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}$ is

$$\begin{aligned} \text{Area}\left(\Delta ABC\right) &= \frac{1}{2}\left|\det\begin{pmatrix} \mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A} \end{pmatrix}\right| \\ &= \frac{1}{2}\left|x_1\left(y_2-y_3\right)+x_2\left(y_3-y_1\right)+x_3\left(y_1-y_2\right)\right| \end{aligned} \tag{1.1.9.1}$$

The area of the quadrilateral $ABCD$, whose diagonal AC lies inside it, is

$$\text{Area}\left(ABCD\right) = \text{Area}\left(\Delta ABC\right)+\text{Area}\left(\Delta ADC\right) \tag{1.1.9.2}$$

Solution: Let $\mathbf{a} = \mathbf{B}-\mathbf{A}$, $\mathbf{b} = \mathbf{C}-\mathbf{A}$ and let θ be the angle at A. Then

$$\text{Area}\left(\Delta ABC\right) = \frac{1}{2}\left\|\mathbf{a}\right\|\left\|\mathbf{b}\right\|\sin\theta \tag{1.1.9.3}$$

Choosing $\mathbf{a} = \left\|\mathbf{a}\right\|\begin{pmatrix} \cos\phi \\ \sin\phi \end{pmatrix}$ and $\mathbf{b} = \left\|\mathbf{b}\right\|\begin{pmatrix} \cos\left(\phi+\theta\right) \\ \sin\left(\phi+\theta\right) \end{pmatrix}$,

$$\det\begin{pmatrix} \mathbf{a} & \mathbf{b} \end{pmatrix} = \left\|\mathbf{a}\right\|\left\|\mathbf{b}\right\|\left[\cos\phi\sin\left(\phi+\theta\right)-\sin\phi\cos\left(\phi+\theta\right)\right] \tag{1.1.9.4}$$

$$= \left\|\mathbf{a}\right\|\left\|\mathbf{b}\right\|\sin\theta \tag{1.1.9.5}$$

Expanding the determinant,

$$\begin{vmatrix} x_2-x_1 & x_3-x_1 \\ y_2-y_1 & y_3-y_1 \end{vmatrix} = x_1\left(y_2-y_3\right)+x_2\left(y_3-y_1\right)+x_3\left(y_1-y_2\right) \tag{1.1.9.6}$$

yielding (1.1.9.1). The modulus accounts for the orientation of the vertices. The diagonal AC splits $ABCD$ into ΔABC and ΔADC, resulting in (1.1.9.2).

1.1.10 Area Ratio Theorem for similar triangles $\Delta ABC \sim \Delta PQR$:

$$\frac{\text{ar}\left(\Delta ABC\right)}{\text{ar}\left(\Delta PQR\right)} = \left(\frac{AB}{PQ}\right)^2 \tag{1.1.10.1}$$

Solution: Similarity with scale factor $s = \frac{PQ}{AB}$ means there exist an orthogonal matrix $\mathbf{M}$, i.e., $\mathbf{M}^\top\mathbf{M} = \mathbf{I}$, with $\left|\mathbf{M}\right| = \pm 1$, and a vector $\mathbf{t}$ such that

$$\mathbf{P} = s\mathbf{M}\mathbf{A}+\mathbf{t}, \quad \mathbf{Q} = s\mathbf{M}\mathbf{B}+\mathbf{t}, \quad \mathbf{R} = s\mathbf{M}\mathbf{C}+\mathbf{t} \tag{1.1.10.2}$$

Then $\mathbf{Q}-\mathbf{P} = s\mathbf{M}\left(\mathbf{B}-\mathbf{A}\right)$ and $\mathbf{R}-\mathbf{P} = s\mathbf{M}\left(\mathbf{C}-\mathbf{A}\right)$. Using (1.1.9.1) and

$$\left|\mathbf{XY}\right| = \left|\mathbf{X}\right|\left|\mathbf{Y}\right|,$$

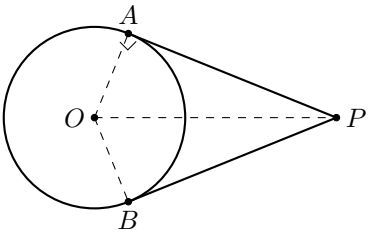
$$\text{ar}\,(\Delta PQR) = \frac{1}{2}\left|\det\left(s\mathbf{M}\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix}\right)\right| \tag{1.1.10.3}$$

$$= \frac{s^2}{2}\left|\det\mathbf{M}\right|\left|\det\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix}\right| = s^2\,\text{ar}\,(\Delta ABC) \tag{1.1.10.4}$$

since $\left|s\mathbf{X}\right| = s^2\left|\mathbf{X}\right|$ for a 2×2 matrix. Hence

$$\frac{\text{ar}\,(\Delta ABC)}{\text{ar}\,(\Delta PQR)} = \frac{1}{s^2} = \left(\frac{AB}{PQ}\right)^2 \tag{1.1.10.5}$$

1.1.11 Tangent properties of a circle with centre $\mathbf{O}$ and radius r:



Radius is perpendicular to the tangent at the point of contact:

$$\angle OAP = 90^\circ \iff (\mathbf{O}-\mathbf{A})^\top(\mathbf{P}-\mathbf{A}) = 0 \tag{1.1.11.1}$$

Tangents drawn to a circle from an external point P are equal in length:

$$PA = PB \tag{1.1.11.2}$$

Solution: The tangent at $\mathbf{A}$ is the line $\mathbf{x} = \mathbf{A}+\lambda\mathbf{m}$. Its points of intersection with the circle $\|\mathbf{x}-\mathbf{O}\|^2 = r^2$ satisfy

$$\|\mathbf{A}-\mathbf{O}+\lambda\mathbf{m}\|^2 = r^2 \implies \|\mathbf{m}\|^2\lambda^2 + 2\lambda\,\mathbf{m}^\top(\mathbf{A}-\mathbf{O}) = 0 \tag{1.1.11.3}$$

using $\|\mathbf{A}-\mathbf{O}\|^2 = r^2$. The roots are $\lambda = 0$ and $\lambda = -\frac{2\mathbf{m}^\top(\mathbf{A}-\mathbf{O})}{\|\mathbf{m}\|^2}$. A tangent touches the circle at only one point, so both roots must be zero, yielding $\mathbf{m}^\top(\mathbf{A}-\mathbf{O}) = 0$, which is (1.1.11.1) since $\mathbf{P}-\mathbf{A}$ is along $\mathbf{m}$.
Now,

$$\|\mathbf{P}-\mathbf{O}\|^2 = \|(\mathbf{P}-\mathbf{A})+(\mathbf{A}-\mathbf{O})\|^2 \tag{1.1.11.4}$$

$$= \|\mathbf{P}-\mathbf{A}\|^2 + 2(\mathbf{P}-\mathbf{A})^\top(\mathbf{A}-\mathbf{O}) + r^2 \tag{1.1.11.5}$$

$$\implies PA^2 = \|\mathbf{P}-\mathbf{O}\|^2 - r^2 \tag{1.1.11.6}$$

by (1.1.11.1). The same steps give $PB^2 = \|\mathbf{P}-\mathbf{O}\|^2 - r^2$, yielding (1.1.11.2).

1.1.12 Pythagoras Theorem in a right-angled triangle ABC with $\angle B = 90^\circ$:

$$\text{Hypotenuse}^2 = \text{Base}^2 + \text{Height}^2 \iff \|\mathbf{A}-\mathbf{C}\|^2 = \|\mathbf{A}-\mathbf{B}\|^2 + \|\mathbf{C}-\mathbf{B}\|^2 \tag{1.1.12.1}$$

Solution: The right angle at B means

$$(\mathbf{A}-\mathbf{B})^{\top}(\mathbf{C}-\mathbf{B})=0 \tag{1.1.12.2}$$

Writing $\mathbf{A}-\mathbf{C}=(\mathbf{A}-\mathbf{B})-(\mathbf{C}-\mathbf{B})$,

$$\|\mathbf{A}-\mathbf{C}\|^{2}=\|\mathbf{A}-\mathbf{B}\|^{2}+\|\mathbf{C}-\mathbf{B}\|^{2}-2(\mathbf{A}-\mathbf{B})^{\top}(\mathbf{C}-\mathbf{B}) \tag{1.1.12.3}$$

yielding (1.1.12.1).

1.1.13 Fundamental trigonometric identities and standard ratios:

$$\sin^{2}A+\cos^{2}A=1 \tag{1.1.13.1}$$
$$\sec^{2}A-\tan^{2}A=1 \tag{1.1.13.2}$$
$$\csc^{2}A-\cot^{2}A=1 \tag{1.1.13.3}$$

For a vector $\mathbf{x}=\begin{pmatrix}\text{Adjacent}\\ \text{Opposite}\end{pmatrix}$ making an angle θ with $\mathbf{e}_1$,

$$\begin{aligned}\sin\theta&=\frac{\text{Opposite}}{\text{Hypotenuse}}=\frac{\mathbf{e}_2^{\top}\mathbf{x}}{\|\mathbf{x}\|}\\ \cos\theta&=\frac{\text{Adjacent}}{\text{Hypotenuse}}=\frac{\mathbf{e}_1^{\top}\mathbf{x}}{\|\mathbf{x}\|}\\ \tan\theta&=\frac{\text{Opposite}}{\text{Adjacent}}=\frac{\mathbf{e}_2^{\top}\mathbf{x}}{\mathbf{e}_1^{\top}\mathbf{x}}\end{aligned} \tag{1.1.13.4}$$

Solution: The point $\mathbf{r}(A)=\begin{pmatrix}\cos A\\ \sin A\end{pmatrix}$ lies on the unit circle, so

$$\|\mathbf{r}(A)\|^{2}=\mathbf{r}(A)^{\top}\mathbf{r}(A)=\cos^{2}A+\sin^{2}A=1 \tag{1.1.13.5}$$

yielding (1.1.13.1). Dividing by $\cos^{2}A$ and $\sin^{2}A$ respectively,

$$1+\tan^{2}A=\sec^{2}A,\quad \cot^{2}A+1=\csc^{2}A \tag{1.1.13.6}$$

which are (1.1.13.2) and (1.1.13.3).

1.1.14 Complementary angle identities:

$$\begin{aligned}\cos(90^{\circ}-\theta)&=\sin\theta\\ \tan(90^{\circ}-\theta)&=\cot\theta\end{aligned} \tag{1.1.14.1}$$

Solution: Rotating the unit vector at angle $-\theta$ by 90° using the rotation matrix,

$$\begin{pmatrix}\cos(90^{\circ}-\theta)\\ \sin(90^{\circ}-\theta)\end{pmatrix}=\begin{pmatrix}0&-1\\ 1&0\end{pmatrix}\begin{pmatrix}\cos\theta\\ -\sin\theta\end{pmatrix}=\begin{pmatrix}\sin\theta\\ \cos\theta\end{pmatrix} \tag{1.1.14.2}$$

yielding the first identity. Dividing the second row by the first,

$$\tan(90^{\circ}-\theta)=\frac{\cos\theta}{\sin\theta}=\cot\theta \tag{1.1.14.3}$$

1.1.15 Area of a circular sector with central angle θ:

$$\text{Area} = \frac{\theta}{360^\circ} \times \pi r^2 \tag{1.1.15.1}$$

Solution: In polar coordinates, for the angle $\theta_r = \frac{\pi\theta}{180^\circ}$ in radians,

$$\text{Area} = \int_0^{\theta_r} \int_0^r \rho \, d\rho \, d\phi = \frac{\theta_r r^2}{2} = \frac{\theta}{360^\circ} \pi r^2 \tag{1.1.15.2}$$

yielding (1.1.15.1). Four 90° corner quadrants combine to form one full circle, since

$$4 \times \frac{90^\circ}{360^\circ} \pi r^2 = \pi r^2 \tag{1.1.15.3}$$

1.1.16 Curved surface areas of a cylinder and a hemisphere of radius r:

$$\text{CSA of cylinder} = 2\pi r h \tag{1.1.16.1}$$

$$\text{CSA of hemisphere} = 2\pi r^2 \tag{1.1.16.2}$$

Solution: For the cylinder, the surface $\mathbf{x} = \begin{pmatrix} r\cos\phi \\ r\sin\phi \\ z \end{pmatrix}$ has area element $r \, d\phi \, dz$,

$$\text{CSA} = \int_0^h \int_0^{2\pi} r \, d\phi \, dz = 2\pi r h \tag{1.1.16.3}$$

For the hemisphere, the area element is $r^2 \sin\alpha \, d\alpha \, d\phi$,

$$\text{CSA} = \int_0^{2\pi} \int_0^{\pi/2} r^2 \sin\alpha \, d\alpha \, d\phi = 2\pi r^2 \left[-\cos\alpha\right]_0^{\pi/2} = 2\pi r^2 \tag{1.1.16.4}$$

1.1.17 Volume, slant height, and curved surface area of a cone of base radius r and height h:

$$V = \frac{1}{3}\pi r^2 h \tag{1.1.17.1}$$

$$l = \left\| \begin{pmatrix} r \\ h \end{pmatrix} \right\| = \sqrt{r^2 + h^2}, \quad \text{CSA} = \pi r l \tag{1.1.17.2}$$

Solution: By similar triangles, the cross-section at height z from the apex has radius $\frac{rz}{h}$. Integrating the areas of the circular cross-sections,

$$V = \int_0^h \pi \left(\frac{rz}{h}\right)^2 dz = \frac{\pi r^2}{h^2} \left[\frac{z^3}{3}\right]_0^h = \frac{1}{3}\pi r^2 h \tag{1.1.17.3}$$

The slant height is the length of the vector $\begin{pmatrix} r \\ h \end{pmatrix}$. Cutting the cone along a slant line and unrolling it gives a sector of radius l and arc length $2\pi r$, so the angle is $\theta_r = \frac{2\pi r}{l}$. From (1.1.15.1) in radians,

$$\text{CSA} = \frac{1}{2} l^2 \theta_r = \pi r l \tag{1.1.17.4}$$

1.1.18 Slant height and surface area of an open frustum with radii $R > r$ and height h:

$$l = \left\| \begin{pmatrix} R-r \\ h \end{pmatrix} \right\| = \sqrt{h^2 + (R-r)^2} \tag{1.1.18.1}$$

$$\text{Area} = \pi (R+r) l + \pi r^2 \tag{1.1.18.2}$$

Solution: The slant height is the length of the vector $\begin{pmatrix} R-r \\ h \end{pmatrix}$. Let L be the slant height of the full cone of base radius R. The removed cone has radius r and slant height $L-l$. By similar triangles,

$$\frac{r}{R} = \frac{L-l}{L} \implies L = \frac{lR}{R-r} \tag{1.1.18.3}$$

Subtracting the curved surface areas using (1.1.17.2),

$$\text{CSA} = \pi RL - \pi r (L-l) = \pi [(R-r) L + rl] = \pi (R+r) l \tag{1.1.18.4}$$

Adding the circular base of radius r yields (1.1.18.2).

1.1.19 Direct method for the mean and the grouped median formula, for class marks x_i and frequencies f_i:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{\mathbf{f}^\top \mathbf{x}}{\mathbf{f}^\top \mathbf{1}} \tag{1.1.19.1}$$

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h \tag{1.1.19.2}$$

where $N = \sum f_i$, l is the lower limit, h the width and f the frequency of the median class, and cf the cumulative frequency of the class preceding it.

Solution: The mean m minimizes the weighted squared error

$$J(m) = \sum f_i (x_i - m)^2 \implies \frac{dJ}{dm} = -2 \sum f_i (x_i - m) = 0 \implies m = \frac{\sum f_i x_i}{\sum f_i} \tag{1.1.19.3}$$

yielding (1.1.19.1). For the median, assuming the frequency is uniformly spread over the median class, the cumulative frequency is linear in that class,

$$F(x) = cf + \frac{f}{h} (x - l) \tag{1.1.19.4}$$

The median is the value where $F(x) = \frac{N}{2}$,

$$\frac{N}{2} = cf + \frac{f}{h} (x - l) \implies x = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h \tag{1.1.19.5}$$

yielding (1.1.19.2).

1.1.20 The unit step function, and an A.P. with first term a and common difference d written

as a sequence:

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{1.1.20.1}$$

$$x(n) = (a + (n-1)d)\ u(n-1) \tag{1.1.20.2}$$

Thus $x(n) = 0$ for $n < 1$ and $x(n) = a_n$ for $n \geq 1$.

1.1.21 Newton–Raphson iteration to find a root of $f(x) = 0$ from a starting value x_0:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad f'(x_n) \neq 0 \tag{1.1.21.1}$$

Solution: The tangent to $y = f(x)$ at $x = x_n$ is $y = f(x_n) + f'(x_n)(x - x_n)$. It meets the x-axis where $y = 0$, which gives x_{n+1} above.

1.1.22 Probability mass function (PMF) of a discrete random variable X, and of a pair (X, Y):

$$p_X(k) = \Pr(X = k), \quad \sum_k p_X(k) = 1, \quad \Pr(X \in E) = \sum_{k \in E} p_X(k) \tag{1.1.22.1}$$

$$p_{XY}(i,j) = p_X(i)\, p_Y(j) \quad \text{when } X \text{ and } Y \text{ are independent} \tag{1.1.22.2}$$

B. Problems

1.2.1 If $x = 3$ is one root of the quadratic equation $x^2 - 2kx - 6 = 0$, then find the value of k.

Solution: Since $x = 3$ is a root of the given quadratic equation, from (1.1.1.1), substituting $x = 3$ into $x^2 - 2kx - 6 = 0$:

$$(3)^2 - 2k(3) - 6 = 0 \tag{1.2.1.1}$$

$$\implies 9 - 6k - 6 = 0 \tag{1.2.1.2}$$

$$\implies 3 - 6k = 0 \tag{1.2.1.3}$$

$$\implies 6k = 3 \tag{1.2.1.4}$$

Thus, the value of k is $\frac{1}{2}$. The following code solves for k, checks that $x = 3$ is a root and plots Fig. 1.2.1.1.

codes/1.2.1.py

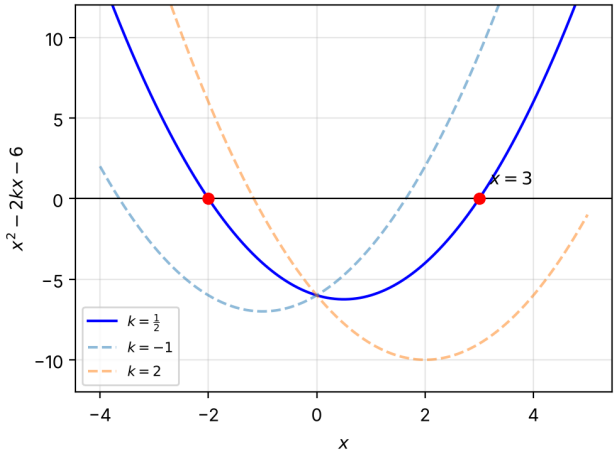


Fig. 1.2.1.1: The parabola $x^2 - 2kx - 6$ for $k = \frac{1}{2}$ crosses the x-axis at $x = 3$

1.2.2 What is the HCF of smallest prime number and the smallest composite number?
Solution:

$$\text{Smallest prime number} = 2 \tag{1.2.2.1}$$

$$\text{Smallest composite number} = 4 \tag{1.2.2.2}$$

Expressing both as prime factors:

$$2 = 2^1 \tag{1.2.2.3}$$

$$4 = 2^2 \tag{1.2.2.4}$$

$$\implies \text{HCF}(2, 4) = 2^1 = 2 \tag{1.2.2.5}$$

1.2.3 Find the distance of a point $P(x, y)$ from the origin.
Solution: From (1.1.7.1), the distance of $P(x, y)$ from $O(0, 0)$ is:

$$d = \sqrt{(x-0)^2 + (y-0)^2} \tag{1.2.3.1}$$

$$= \sqrt{x^2 + y^2} \tag{1.2.3.2}$$

1.2.4 In an AP, if the common difference $(d) = -4$, and the seventh term (a_7) is 4, then find the first term.
Solution: The A.P. is the sequence $x(n)$ with first term a and common difference $d = -4$. From (1.1.20.2),

$$x(n) = (a + (n-1)d)\ u(n-1) = (a - 4(n-1))\ u(n-1) \tag{1.2.4.1}$$

(1.1.20.2)

Since $u(6) = 1$, the seventh term is $x(7) = a - 24$. Equating it to the given value,

$$a - 24 = 4 \implies a = 28 \tag{1.2.4.2}$$

The first term is therefore $\boxed{28}$. The C program prints the sequence, which the unit step switches on at $n = 1$, and the Python program plots it in Fig. 1.2.4.1.

codes/1.2.4.c

codes/1.2.4.py

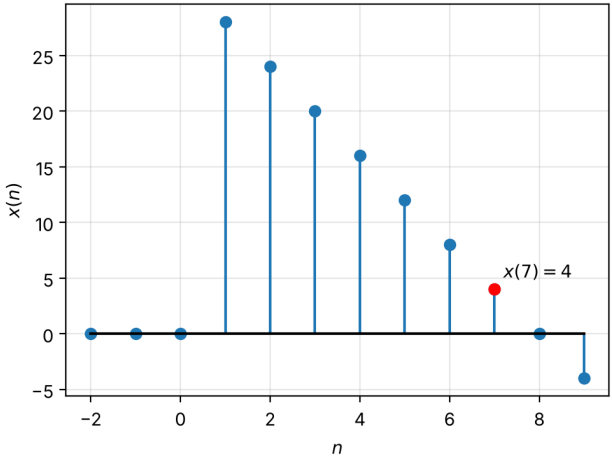


Fig. 1.2.4.1: The A.P. as the sequence $x(n) = (a + (n-1)d)\,u(n-1)$

1.2.5 What is the value of $(\cos^2 67° - \sin^2 23°)$?
Solution: From (1.1.14.1), $\cos(90° - \theta) = \sin\theta$:

$$\cos 67° = \cos(90° - 23°) \tag{1.2.5.1}$$

$$\implies \cos 67° = \sin 23° \tag{1.2.5.2}$$

$$\implies \cos^2 67° - \sin^2 23° = \sin^2 23° - \sin^2 23° = 0 \tag{1.2.5.3}$$

1.2.6 Given $\Delta ABC \sim \Delta PQR$, if $\frac{AB}{PQ} = \frac{1}{3}$, then find $\frac{\text{ar}(\Delta ABC)}{\text{ar}(\Delta PQR)}$.
Solution: From (1.1.10.1) (area ratio theorem for similar triangles):

$$\frac{\text{ar}(\Delta ABC)}{\text{ar}(\Delta PQR)} = \left(\frac{AB}{PQ}\right)^2 \tag{1.2.6.1}$$

$$\implies \frac{\text{ar}(\Delta ABC)}{\text{ar}(\Delta PQR)} = \left(\frac{1}{3}\right)^2 = \frac{1}{9} \tag{1.2.6.2}$$

1.2.7 Given that $\sqrt{2}$ is irrational, prove that $(5 + 3\sqrt{2})$ is an irrational number.

Solution: Assume $(5+3\sqrt{2})$ is rational, so $5+3\sqrt{2}=\frac{a}{b}$ where $a,b\in\mathbb{Z},b\neq 0$:

$$3\sqrt{2}=\frac{a}{b}-5 \tag{1.2.7.1}$$

$$\implies \sqrt{2}=\frac{a-5b}{3b} \tag{1.2.7.2}$$

Since $a,b\in\mathbb{Z}$, the expression $\frac{a-5b}{3b}$ is rational. This contradicts the given fact that $\sqrt{2}$ is irrational. Hence, $(5+3\sqrt{2})$ is irrational.

1.2.8 In Fig. 1.2.8.1, $ABCD$ is a rectangle. Find the values of x and y.

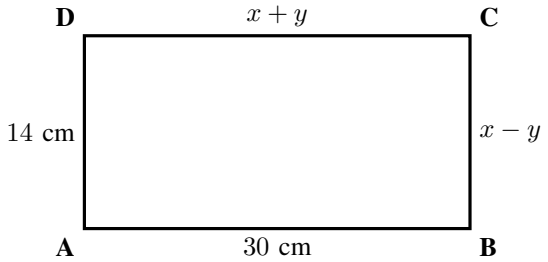


Fig. 1.2.8.1: Rectangle $ABCD$

Solution: Opposite sides of a rectangle are equal in length.

$$x+y=30 \tag{1.2.8.1}$$

$$x-y=14 \tag{1.2.8.2}$$

Adding (1.2.8.1) and (1.2.8.2):

$$(x+y)+(x-y)=30+14 \tag{1.2.8.3}$$

$$\implies 2x=44 \tag{1.2.8.4}$$

$$\implies x=22 \tag{1.2.8.5}$$

Substituting $x=22$ into (1.2.8.1):

$$22+y=30 \tag{1.2.8.6}$$

$$\implies y=30-22 \tag{1.2.8.7}$$

$$\implies y=8 \tag{1.2.8.8}$$

Thus, $x=22$ and $y=8$.

1.2.9 Find the sum of first 8 multiples of 3.

Solution: The first 8 multiples of 3 form an A.P.: $3,6,9,\ldots$ with first term $a=3$,

common difference $d = 3$, and $n = 8$. From (1.1.5.1),

$$S_n = \frac{n}{2}[2a + (n-1)d] \tag{1.2.9.1}$$

$$\implies S_8 = \frac{8}{2}[2(3) + (8-1)3] \tag{1.2.9.2}$$

$$\implies S_8 = 4[6 + 21] \tag{1.2.9.3}$$

$$\implies S_8 = 4(27) = 108 \tag{1.2.9.4}$$

1.2.10 Find the ratio in which $P(4, m)$ divides the line segment joining the points $A(2, 3)$ and $B(6, -3)$. Hence find m.

Solution: Let $P(4, m)$ divide AB in the ratio $k : 1$. From (1.1.8.1), for the x-coordinate:

$$x = \frac{k(x_2) + 1(x_1)}{k+1} \tag{1.2.10.1}$$

$$\implies 4 = \frac{k(6) + 1(2)}{k+1} \tag{1.2.10.2}$$

$$\implies 4(k+1) = 6k + 2 \tag{1.2.10.3}$$

$$\implies 4k + 4 = 6k + 2 \tag{1.2.10.4}$$

$$\implies 2k = 2 \implies k = 1 \tag{1.2.10.5}$$

Thus, the ratio is $1 : 1$. Now from (1.1.8.1), for the y-coordinate:

$$m = \frac{1(-3) + 1(3)}{1+1} \tag{1.2.10.6}$$

$$\implies m = \frac{0}{2} = 0 \tag{1.2.10.7}$$

The following code verifies the result and generates Fig. 1.2.10.1.

codes/1.2.10.py

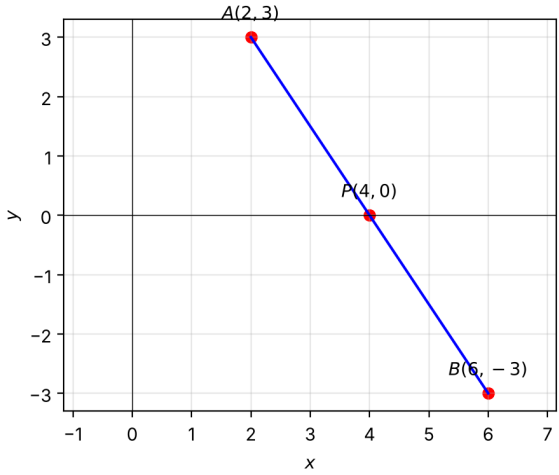


Fig. 1.2.10.1: Point P dividing AB

1.2.11 Two different dice are tossed together. Find the probability:

(i) of getting a doublet.

(ii) of getting a sum 10, of the numbers on the two dice.

Solution: Let X and Y be the numbers on the two dice. They are independent and each is uniform on $\{1,\dots,6\}$, so from (2.1.20.2),

$$p_{XY}(i,j) = \frac{1}{6}\cdot\frac{1}{6} = \frac{1}{36}, \quad i,j \in \{1,\dots,6\} \tag{1.2.11.1}$$

(i) A doublet means $X = Y$. From (2.1.20.1),

$$\Pr(X=Y) = \sum_{i=1}^{6} p_{XY}(i,i) = 6\cdot\frac{1}{36} = \frac{1}{6} \tag{1.2.11.2}$$

(ii) A sum of 10 means $(X,Y) \in \{(4,6),(5,5),(6,4)\}$:

$$\Pr(X+Y=10) = \sum_{i=4}^{6} p_{XY}(i,10-i) = 3\cdot\frac{1}{36} = \frac{1}{12} \tag{1.2.11.3}$$

The following code builds the joint PMF of the two dice and verifies both probabilities.

codes/1.2.11.py

1.2.12 An integer is chosen at random between 1 and 100. Find the probability that it is:

(i) divisible by 8.

(ii) not divisible by 8.

Solution: Taking "between 1 and 100" to exclude both 1 and 100, the integer X

takes the values $2, 3, \ldots, 99$, which are 98 equally likely values. Its PMF is

$$p_X(k) = \frac{1}{98}, \quad k \in \{2, 3, \ldots, 99\} \tag{1.2.12.1}$$

(i) The multiples of 8 are $E = \{8, 16, \ldots, 96\}$, which has 12 values. From (2.1.20.1),

$$\Pr(8 \mid X) = \sum_{k \in E} p_X(k) = \frac{12}{98} = \frac{6}{49} \tag{1.2.12.2}$$

(ii) Since $\sum_k p_X(k) = 1$,

$$\Pr(8 \nmid X) = 1 - \sum_{k \in E} p_X(k) = 1 - \frac{6}{49} = \frac{43}{49} \tag{1.2.12.3}$$

1.2.13 Find HCF and LCM of 404 and 96 and verify that HCF $\times$ LCM $=$ Product of the two given numbers.

Solution: Expressing numbers as products of prime factors:

$$404 = 2^2 \times 101 \tag{1.2.13.1}$$
$$96 = 2^5 \times 3 \tag{1.2.13.2}$$
$$\implies \text{HCF}(404, 96) = 2^2 = 4 \tag{1.2.13.3}$$
$$\implies \text{LCM}(404, 96) = 2^5 \times 3 \times 101 = 9696 \tag{1.2.13.4}$$

Verification:

$$\text{LHS} = \text{HCF} \times \text{LCM} = 4 \times 9696 = 38784 \tag{1.2.13.5}$$
$$\text{RHS} = 404 \times 96 = 38784 \tag{1.2.13.6}$$
$$\implies \text{LHS} = \text{RHS} \tag{1.2.13.7}$$

1.2.14 Find all zeroes of the polynomial $(2x^4 - 9x^3 + 5x^2 + 3x - 1)$ if two of its zeroes are $(2 + \sqrt{3})$ and $(2 - \sqrt{3})$.

Solution: Given zeroes $x = 2 \pm \sqrt{3}$, from (1.1.2.1), $(x - \alpha)(x - \beta)$ is a factor of $p(x)$:

$$[x - (2 + \sqrt{3})][x - (2 - \sqrt{3})] = (x - 2)^2 - 3 \tag{1.2.14.1}$$
$$= x^2 - 4x + 4 - 3 \tag{1.2.14.2}$$
$$\implies g(x) = x^2 - 4x + 1 \text{ is a factor} \tag{1.2.14.3}$$

Dividing $p(x) = 2x^4 - 9x^3 + 5x^2 + 3x - 1$ by $g(x) = x^2 - 4x + 1$:

$$2x^4 - 9x^3 + 5x^2 + 3x - 1 = (x^2 - 4x + 1)(2x^2 - x - 1) \tag{1.2.14.4}$$

To find the remaining zeroes, set $2x^2 - x - 1 = 0$:

$$2x^2 - 2x + x - 1 = 0 \tag{1.2.14.5}$$
$$2x(x-1) + 1(x-1) = 0 \tag{1.2.14.6}$$
$$(2x+1)(x-1) = 0 \tag{1.2.14.7}$$
$$\implies x = 1 \quad \text{or} \quad x = -\frac{1}{2} \tag{1.2.14.8}$$

Thus, all zeroes are $2+\sqrt{3}$, $2-\sqrt{3}$, 1, and $-\frac{1}{2}$.The following code verifies the zeroes and plots $p(x)$ in Fig. 1.2.14.1.

codes/1.2.14.py

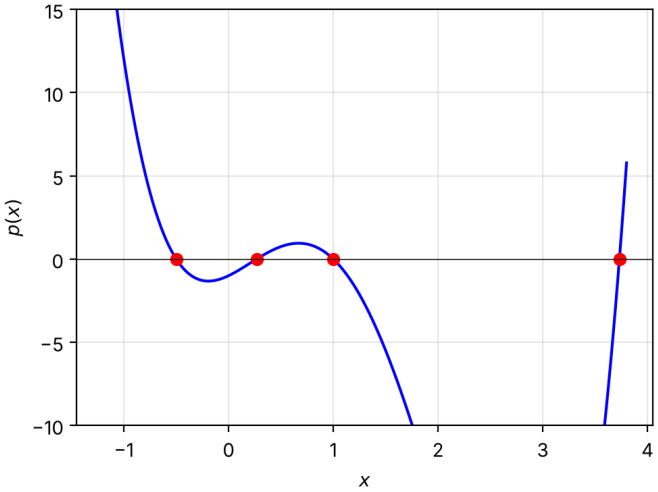


Fig. 1.2.14.1: Plot of $p(x) = 2x^4 - 9x^3 + 5x^2 + 3x - 1$ with its zeroes

1.2.15 (a) If $A(-2,1)$, $B(a,0)$, $C(4,b)$ and $D(1,2)$ are the vertices of a parallelogram $ABCD$, find the values of a and b. Hence find the lengths of its sides.

OR

(b) If $A(-5,7)$, $B(-4,-5)$, $C(-1,-6)$ and $D(4,5)$ are the vertices of a quadrilateral, find the area of the quadrilateral $ABCD$.

(a) Solution: In a parallelogram, the diagonals bisect each other. Therefore, the midpoint of diagonal AC is equal to the midpoint of diagonal BD.
From (1.1.8.1) with $k = 1$, the midpoint of AC is:

$$M_{AC} = \left(\frac{-2+4}{2}, \frac{1+b}{2}\right) = \left(1, \frac{1+b}{2}\right) \tag{1.2.15.1}$$

Midpoint of BD:

$$M_{BD} = \left(\frac{a+1}{2}, \frac{0+2}{2}\right) = \left(\frac{a+1}{2}, 1\right) \tag{1.2.15.2}$$

Equating the x-coordinates and y-coordinates:

$$\frac{a+1}{2} = 1 \implies a+1 = 2 \implies a = 1 \tag{1.2.15.3}$$

$$\frac{1+b}{2} = 1 \implies 1+b = 2 \implies b = 1 \tag{1.2.15.4}$$

Thus, the vertices are $A(-2,1)$, $B(1,0)$, $C(4,1)$, and $D(1,2)$.

Now, from (1.1.7.1), the lengths of the sides are:

$$AB = \sqrt{(1-(-2))^2 + (0-1)^2} = \sqrt{3^2 + (-1)^2} = \sqrt{9+1} = \sqrt{10} \tag{1.2.15.5}$$

$$BC = \sqrt{(4-1)^2 + (1-0)^2} = \sqrt{3^2 + 1^2} = \sqrt{9+1} = \sqrt{10} \tag{1.2.15.6}$$

Since opposite sides of a parallelogram are equal:

$$CD = AB = \sqrt{10} \tag{1.2.15.7}$$

$$AD = BC = \sqrt{10} \tag{1.2.15.8}$$

Thus, $a = 1, b = 1$, and all four sides have length $\sqrt{10}$ units.

(b) Solution: Joining diagonal AC divides quadrilateral $ABCD$ into two triangles: ΔABC and ΔADC.
From (1.1.9.1), the area of a triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ is given by:

$$\text{Area} = \frac{1}{2}\left|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right| \tag{1.2.15.9}$$

For ΔABC with $A(-5,7)$, $B(-4,-5)$, and $C(-1,-6)$:

$$\text{Area}(\Delta ABC) = \frac{1}{2}\left| -5(-5-(-6)) + (-4)(-6-7) + (-1)(7-(-5))\right| \tag{1.2.15.10}$$

$$= \frac{1}{2}\left| -5(1) + (-4)(-13) + (-1)(12)\right| \tag{1.2.15.11}$$

$$= \frac{1}{2}\left| -5 + 52 - 12\right| \tag{1.2.15.12}$$

$$= \frac{1}{2}|35| = 17.5 \text{ sq. units} \tag{1.2.15.13}$$

For ΔADC with $A(-5,7)$, $D(4,5)$, and $C(-1,-6)$:

$$\text{Area}(\Delta ADC) = \frac{1}{2}\left|-5(5-(-6))+4(-6-7)+(-1)(7-5)\right| \tag{1.2.15.14}$$

$$= \frac{1}{2}\left|-5(11)+4(-13)+(-1)(2)\right| \tag{1.2.15.15}$$

$$= \frac{1}{2}\left|-55-52-2\right| \tag{1.2.15.16}$$

$$= \frac{1}{2}\left|-109\right| = 54.5 \text{ sq. units} \tag{1.2.15.17}$$

From (1.1.9.2), the total area of quadrilateral $ABCD$ is:

$$\text{Area}(ABCD) = \text{Area}(\Delta ABC) + \text{Area}(\Delta ADC) \tag{1.2.15.18}$$

$$= 17.5 + 54.5 = 72 \text{ sq. units} \tag{1.2.15.19}$$

Thus, the area of quadrilateral $ABCD$ is 72 sq. units.The following code verifies both parts and generates Fig. 1.2.15.1.

codes/1.2.15.py

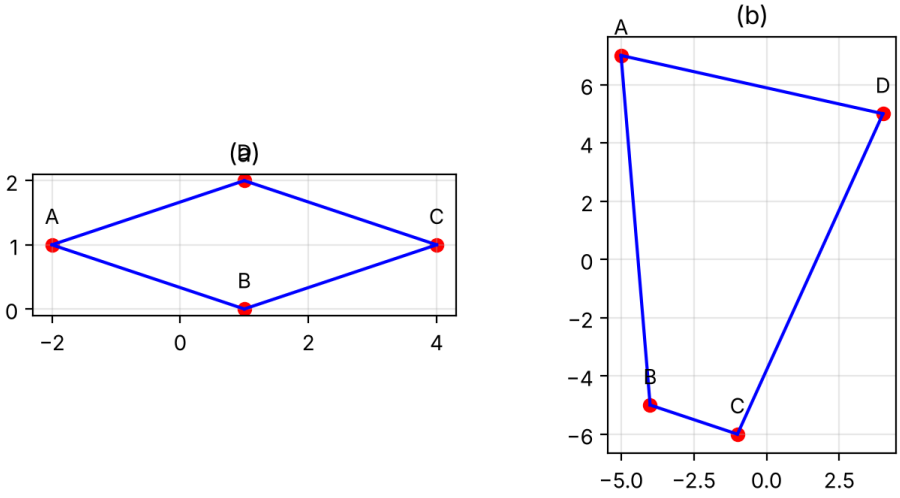


Fig. 1.2.15.1: (a) Parallelogram $ABCD$ and (b) quadrilateral $ABCD$

1.2.16 A plane left 30 minutes late than its scheduled time and in order to reach the destination 1500 km away in time, it had to increase its speed by 100 km/h from the usual speed. Find its usual speed.

Solution: Let the usual speed be x km/h. The delay of $\frac{1}{2}$ hour on 1500 km gives

$$\frac{1500}{x} - \frac{1500}{x+100} = \frac{1}{2} \implies f(x) = x^2 + 100x - 300000 = 0 \tag{1.2.16.1}$$

The speed is the positive root of f. With $f'(x) = 2x + 100$, the Newton–Raphson iteration is

$$x_{n+1} = x_n - \frac{x_n^2 + 100x_n - 300000}{2x_n + 100} \tag{1.2.16.2}$$

(1.1.21.1)

Starting from $x_0 = 1000$ and from $x_0 = -1000$, the C program gives

n	x_n $(x_0 = 1000)$	x_n $(x_0 = -1000)$
0	1000.000000	−1000.000000
1	619.047619	−684.210526
2	510.591425	−605.590740
3	500.100054	−600.028129
4	500.000009	−600.000001
5	500.000000	−600.000000
6	500.000000	−600.000000

The iterates settle at 500 and -600. Speed cannot be negative, so the usual speed is $\boxed{500 \text{ km/h}}$. This agrees with $f(x) = (x - 500)(x + 600)$. The C program prints the iterates and checks both roots, and the Python program draws the first three Newton steps as tangent lines using CoordGeo in Fig. 1.2.16.1.

codes/1.2.16.c

codes/1.2.16.py

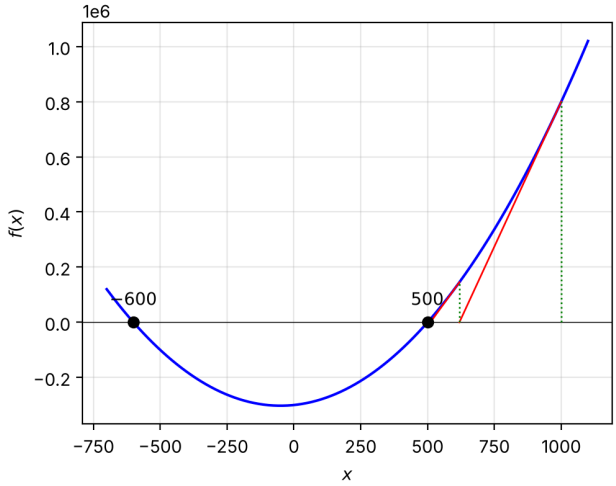


Fig. 1.2.16.1: Newton–Raphson steps for $f(x) = x^2 + 100x - 300000$ from $x_0 = 1000$

1.2.17 (a) Prove that the area of an equilateral triangle described on one side of a square is equal to half the area of the equilateral triangle described on one of its diagonals.

OR

(b) If the areas of two similar triangles are equal, prove that they are congruent.

(a) Solution: Let $ABCD$ be a square of side length a. From (1.1.12.1), the length of diagonal AC is given by:

$$AC = \sqrt{a^2 + a^2} = a\sqrt{2} \tag{1.2.17.1}$$

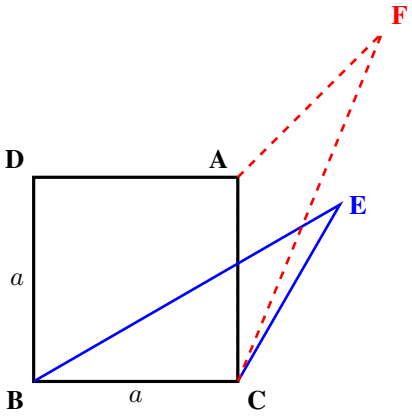


Fig. 1.2.17.1: Equilateral triangles ΔBCE on side a and ΔACF on diagonal $a\sqrt{2}$

Let ΔBCE be the equilateral triangle described on side $BC = a$, and ΔACF be the equilateral triangle described on diagonal $AC = a\sqrt{2}$.
Area of an equilateral triangle with side s is given by $\frac{\sqrt{3}}{4}s^2$.
Area of ΔBCE:

$$\text{Area}(\Delta BCE) = \frac{\sqrt{3}}{4}a^2 \tag{1.2.17.2}$$

Area of ΔACF:

$$\text{Area}(\Delta ACF) = \frac{\sqrt{3}}{4}(a\sqrt{2})^2 = \frac{\sqrt{3}}{4}(2a^2) = 2\left(\frac{\sqrt{3}}{4}a^2\right) \tag{1.2.17.3}$$

$$\implies \text{Area}(\Delta BCE) = \frac{1}{2}\text{Area}(\Delta ACF) \tag{1.2.17.4}$$

Hence proved.

(b) Solution: Let $\Delta ABC \sim \Delta PQR$ such that $\text{Area}(\Delta ABC) = \text{Area}(\Delta PQR)$.
From (1.1.10.1) (area ratio theorem for similar triangles):

$$\frac{\text{Area}(\Delta ABC)}{\text{Area}(\Delta PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{AC}{PR}\right)^2 \tag{1.2.17.5}$$

Since Area(ΔABC) = Area(ΔPQR), the ratio of their areas is 1:

$$1 = \left(\frac{AB}{PQ}\right)^2 \implies \frac{AB}{PQ} = 1 \implies AB = PQ \tag{1.2.17.6}$$

$$1 = \left(\frac{BC}{QR}\right)^2 \implies \frac{BC}{QR} = 1 \implies BC = QR \tag{1.2.17.7}$$

$$1 = \left(\frac{AC}{PR}\right)^2 \implies \frac{AC}{PR} = 1 \implies AC = PR \tag{1.2.17.8}$$

Since all three corresponding sides are equal ($AB = PQ$, $BC = QR$, $AC = PR$), by SSS congruence criterion:

$$\Delta ABC \cong \Delta PQR \tag{1.2.17.9}$$

Hence proved.

1.2.18 Prove that the lengths of tangents drawn from an external point to a circle are equal.

Solution: Let P be an external point to a circle with centre O. Let PA and PB be two tangents drawn from P touching the circle at points A and B respectively. Join OA, OB, and OP.

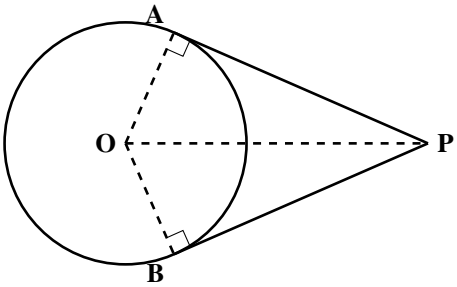


Fig. 1.2.18.1: Tangents PA and PB from external point P to circle with centre O

In ΔOAP and ΔOBP:

$$\angle OAP = \angle OBP = 90^\circ \quad \text{(Radius is perpendicular to tangent, from (1.1.11.1))} \tag{1.2.18.1}$$

$$OP = OP \quad \text{(Common hypotenuse)} \tag{1.2.18.2}$$

$$OA = OB \quad \text{(Radii of the same circle)} \tag{1.2.18.3}$$

By RHS congruence criterion:

$$\Delta OAP \cong \Delta OBP \implies PA = PB \tag{1.2.18.4}$$

Hence, the lengths of tangents drawn from an external point to a circle are equal.

1.2.19 (a) If $4\tan\theta = 3$, evaluate $\left(\frac{4\sin\theta - \cos\theta + 1}{4\sin\theta + \cos\theta - 1}\right)$.

OR

(b) If $\tan 2A = \cot(A - 18^\circ)$, where $2A$ is an acute angle, find the value of A.

(a) Solution: Given $4\tan\theta = 3 \implies \tan\theta = \frac{3}{4}$.
Consider a right-angled triangle with opposite side $= 3$ and adjacent side $= 4$. By Pythagoras theorem, from (1.1.12.1), the hypotenuse is:

$$\text{Hypotenuse} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5 \tag{1.2.19.1}$$

Therefore, from (1.1.13.4):

$$\sin\theta = \frac{3}{5}, \quad \cos\theta = \frac{4}{5} \tag{1.2.19.2}$$

Substituting these values into the given expression:

$$\frac{4\sin\theta - \cos\theta + 1}{4\sin\theta + \cos\theta - 1} = \frac{4\left(\frac{3}{5}\right) - \frac{4}{5} + 1}{4\left(\frac{3}{5}\right) + \frac{4}{5} - 1} \tag{1.2.19.3}$$

$$= \frac{\frac{12}{5} - \frac{4}{5} + 1}{\frac{12}{5} + \frac{4}{5} - 1} \tag{1.2.19.4}$$

$$= \frac{\frac{8}{5} + 1}{\frac{16}{5} - 1} \tag{1.2.19.5}$$

$$= \frac{\frac{13}{5}}{\frac{11}{5}} = \frac{13}{11} \tag{1.2.19.6}$$

Thus, the value of the expression is $\frac{13}{11}$.

(b) Solution: Given $\tan 2A = \cot(A - 18^\circ)$.
From (1.1.14.1), $\tan\theta = \cot(90^\circ - \theta)$:

$$\cot(90^\circ - 2A) = \cot(A - 18^\circ) \tag{1.2.19.7}$$

Equating the angles:

$$90^\circ - 2A = A - 18^\circ \tag{1.2.19.8}$$

$$\implies 90^\circ + 18^\circ = A + 2A \tag{1.2.19.9}$$

$$\implies 3A = 108^\circ \tag{1.2.19.10}$$

$$\implies A = \frac{108^\circ}{3} = 36^\circ \tag{1.2.19.11}$$

Thus, the value of A is 36°.

1.2.20 Find the area of the shaded region in Fig. 1.2.20.1, where arcs drawn with centres A, B, C and D intersect in pairs at mid-points P, Q, R and S of the sides AB, BC, CD and DA respectively of a square $ABCD$ of side 12 cm. [Use $\pi = 3.14$]

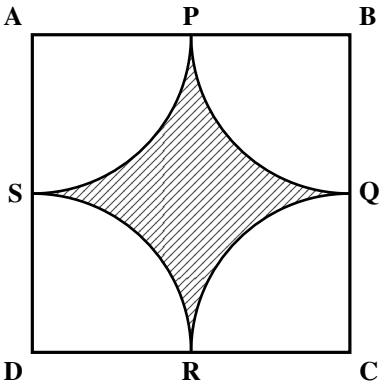


Fig. 1.2.20.1: Shaded region enclosed by circular arcs in square $ABCD$

Solution: Given:

$$\text{Side of square } ABCD \ (s) = 12 \text{ cm} \tag{1.2.20.1}$$

Since P, Q, R, S are the mid-points of the sides of the square, the radius r of each corner sector is:

$$r = \frac{12}{2} = 6 \text{ cm} \tag{1.2.20.2}$$

Each corner sector has an angle of 90° (since all interior angles of a square are 90°), making each sector a quadrant of a circle.
From (1.1.15.1), combining these 4 identical quadrants forms 1 complete circle of radius $r = 6$ cm:

$$\text{Area of 4 quadrants} = \text{Area of 1 full circle} = \pi r^2 \tag{1.2.20.3}$$

$$= 3.14 \times (6)^2 = 3.14 \times 36 = 113.04 \text{ cm}^2 \tag{1.2.20.4}$$

Calculating the area of the square:

$$\text{Area of square } ABCD = s^2 = (12)^2 = 144 \text{ cm}^2 \tag{1.2.20.5}$$

Subtracting the area of the circle from the area of the square gives the area of the shaded region:

$$\text{Area of shaded region} = \text{Area of square } ABCD - \text{Area of 4 quadrants} \tag{1.2.20.6}$$

$$= 144 - 113.04 = 30.96 \text{ cm}^2 \tag{1.2.20.7}$$

Thus, the area of the shaded region is 30.96 cm^2.

1.2.21 (a) A wooden article was made by scooping out a hemisphere from each end of a solid cylinder, as shown in Fig. 1.2.21.1. If the height of the cylinder is 10 cm and its base is of radius 3.5 cm, find the total surface area of the article.

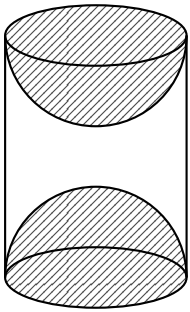


Fig. 1.2.21.1: Wooden article

OR

(b) A heap of rice is in the form of a cone of base diameter 24 m and height 3.5 m. Find the volume of the rice. How much canvas cloth is required to just cover the heap?

(a) Solution: Given radius $r = 3.5$ cm $= \frac{7}{2}$ cm and height of cylinder $h = 10$ cm.

From (1.1.16.1) and (1.1.16.2), the total surface area (TSA) of the article consists of the curved surface area (CSA) of the cylinder plus the CSA of two hemispherical ends:

$$\begin{aligned} \text{TSA} &= \text{CSA of cylinder} + 2 \times (\text{CSA of hemisphere}) && (1.2.21.1)\\ &= 2\pi rh + 2 \times (2\pi r^2) && (1.2.21.2)\\ &= 2\pi rh + 4\pi r^2 && (1.2.21.3)\\ &= 2\pi r(h + 2r) && (1.2.21.4) \end{aligned}$$

Substituting $r = \frac{7}{2}$ and $h = 10$:

$$\begin{aligned} \text{TSA} &= 2 \times \frac{22}{7} \times \frac{7}{2}\left(10 + 2 \times \frac{7}{2}\right) && (1.2.21.5)\\ &= 22 \times (10 + 7) && (1.2.21.6)\\ &= 22 \times 17 = 374 \text{ cm}^2 && (1.2.21.7) \end{aligned}$$

Thus, the total surface area of the article is 374 cm^2.

(b) Solution: Given diameter $d = 24$ m $\implies$ radius $r = 12$ m, and height $h = 3.5$ m $= \frac{7}{2}$ m.

From (1.1.17.1), the volume of the conical heap of rice is:

$$V = \frac{1}{3}\pi r^2 h \tag{1.2.21.8}$$
$$= \frac{1}{3} \times \frac{22}{7} \times (12)^2 \times \frac{7}{2} \tag{1.2.21.9}$$
$$= \frac{1}{3} \times \frac{22}{7} \times 144 \times \frac{7}{2} \tag{1.2.21.10}$$
$$= 11 \times 48 = 528 \text{ m}^3 \tag{1.2.21.11}$$

To find the canvas cloth required to cover the heap, from (1.1.17.2) we calculate the curved surface area (CSA $= \pi rl$), where l is the slant height:

$$l = \sqrt{r^2 + h^2} = \sqrt{12^2 + (3.5)^2} \tag{1.2.21.12}$$
$$= \sqrt{144 + 12.25} = \sqrt{156.25} = 12.5 \text{ m} \tag{1.2.21.13}$$

Area of canvas cloth required:

$$\text{CSA} = \pi rl \tag{1.2.21.14}$$
$$= \frac{22}{7} \times 12 \times 12.5 \tag{1.2.21.15}$$
$$= \frac{22 \times 150}{7} = \frac{3300}{7} \approx 471.43 \text{ m}^2 \tag{1.2.21.16}$$

Thus, the volume of rice is 528 m^3 and the area of canvas cloth required is 471.43 m^2.

1.2.22 The table below shows the salaries of 280 persons:

Salary (in thousand Rs.)	**No. of Persons**
5–10	49
10–15	133
15–20	63
20–25	15
25–30	6
30–35	7
35–40	4
40–45	2
45–50	1

Calculate the median salary of the data.

Solution: Constructing the cumulative frequency table:

Salary (in thousand Rs.)	Frequency (f)	Cumulative Frequency (cf)
5–10	49	49
10–15	133	182
15–20	63	245
20–25	15	260
25–30	6	266
30–35	7	273
35–40	4	277
40–45	2	279
45–50	1	280

Here, $N = 280 \implies \frac{N}{2} = 140$.

The cumulative frequency just greater than 140 is 182, which corresponds to the median class 10–15.

Lower limit of median class $l = 10$, $cf = 49$, $f = 133$, and class width $h = 5$. From (1.1.19.2),

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h \tag{1.2.22.1}$$

$$= 10 + \left(\frac{140 - 49}{133}\right) \times 5 \tag{1.2.22.2}$$

$$= 10 + \left(\frac{91}{133}\right) \times 5 \tag{1.2.22.3}$$

$$= 10 + \frac{455}{133} \approx 10 + 3.42 = 13.42 \tag{1.2.22.4}$$

Thus, the median salary is Rs. 13.42 thousand = Rs. $13,420$.The following code computes the median and plots the ogive in Fig. 1.2.22.1.

codes/1.2.22.py

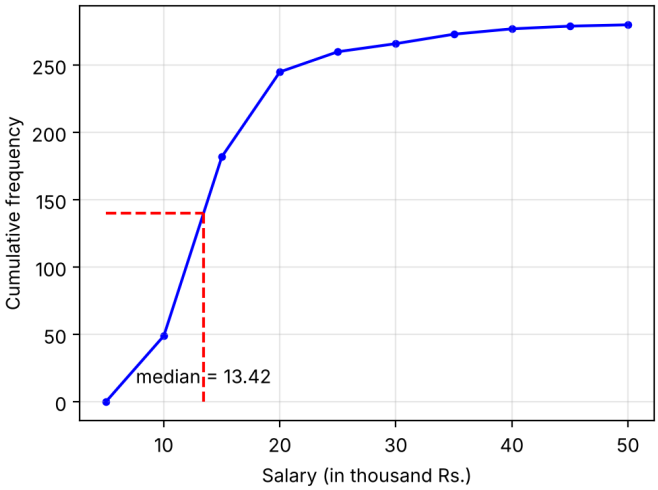


Fig. 1.2.22.1: Less than type ogive for the salary data

1.2.23 (a) A motor boat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

OR

(b) A train travels at a certain average speed for a distance of 63 km and then travels at a distance of 72 km at an average speed of 6 km/h more than its original speed. If it takes 3 hours to complete total journey, what is the original average speed?

(a) Solution: Let the speed of the stream be x km/h.
Given:

$$\text{Speed of boat in still water} = 18 \text{ km/h} \tag{1.2.23.1}$$

$$\text{Speed upstream} = (18 - x) \text{ km/h} \tag{1.2.23.2}$$

$$\text{Speed downstream} = (18 + x) \text{ km/h} \tag{1.2.23.3}$$

$$\text{Distance} = 24 \text{ km} \tag{1.2.23.4}$$

Time taken upstream $t_1 = \frac{24}{18-x}$ hours, and time taken downstream $t_2 = \frac{24}{18+x}$ hours.

According to the given condition ($t_1 - t_2 = 1$ hour):

$$\frac{24}{18-x} - \frac{24}{18+x} = 1 \tag{1.2.23.5}$$

$$24\left(\frac{(18+x)-(18-x)}{(18-x)(18+x)}\right) = 1 \tag{1.2.23.6}$$

$$24\left(\frac{2x}{324-x^2}\right) = 1 \tag{1.2.23.7}$$

$$48x = 324 - x^2 \tag{1.2.23.8}$$

$$\implies x^2 + 48x - 324 = 0 \tag{1.2.23.9}$$

Factoring the quadratic equation:

$$x^2 + 54x - 6x - 324 = 0 \tag{1.2.23.10}$$

$$x(x+54) - 6(x+54) = 0 \tag{1.2.23.11}$$

$$(x-6)(x+54) = 0 \tag{1.2.23.12}$$

$$\implies x = 6 \quad \text{or} \quad x = -54 \tag{1.2.23.13}$$

Since speed cannot be negative, $x = 6$. Thus, the speed of the stream is 6 km/h.

(b) Solution: Let the original average speed of the train be x km/h.
Given:

$$\text{Time for first 63 km} = \frac{63}{x} \text{ hours} \tag{1.2.23.14}$$

$$\text{Time for next 72 km} = \frac{72}{x+6} \text{ hours} \tag{1.2.23.15}$$

Total journey time is 3 hours:

$$\frac{63}{x} + \frac{72}{x+6} = 3 \tag{1.2.23.16}$$

Dividing the entire equation by 3:

$$\frac{21}{x} + \frac{24}{x+6} = 1 \tag{1.2.23.17}$$

$$\frac{21(x+6)+24x}{x(x+6)} = 1 \tag{1.2.23.18}$$

$$21x + 126 + 24x = x^2 + 6x \tag{1.2.23.19}$$

$$45x + 126 = x^2 + 6x \tag{1.2.23.20}$$

$$\implies x^2 - 39x - 126 = 0 \tag{1.2.23.21}$$

Factoring the quadratic equation:

$$x^2 - 42x + 3x - 126 = 0 \tag{1.2.23.22}$$
$$x(x-42) + 3(x-42) = 0 \tag{1.2.23.23}$$
$$(x-42)(x+3) = 0 \tag{1.2.23.24}$$
$$\implies x = 42 \quad \text{or} \quad x = -3 \tag{1.2.23.25}$$

Since speed cannot be negative, $x = 42$. Thus, the original average speed of the train is 42 km/h.The following code verifies both parts.

codes/1.2.23.py

1.2.24 The sum of four consecutive numbers in an AP is 32 and the ratio of the product of the first and the last term to the product of two middle terms is $7 : 15$. Find the numbers.

Solution: From (1.1.6.1), let the four consecutive terms of the A.P. be $(a-3d), (a-d), (a+d)$, and $(a+3d)$.

Given that the sum of these four numbers is 32:

$$(a-3d) + (a-d) + (a+d) + (a+3d) = 32 \tag{1.2.24.1}$$
$$\implies 4a = 32 \tag{1.2.24.2}$$
$$\implies a = 8 \tag{1.2.24.3}$$

Also given that the ratio of the product of the first and last terms to the product of the middle terms is $7 : 15$:

$$\frac{(a-3d)(a+3d)}{(a-d)(a+d)} = \frac{7}{15} \tag{1.2.24.4}$$
$$\implies \frac{a^2 - 9d^2}{a^2 - d^2} = \frac{7}{15} \tag{1.2.24.5}$$

Substituting $a = 8$ ($a^2 = 64$):

$$\frac{64 - 9d^2}{64 - d^2} = \frac{7}{15} \tag{1.2.24.6}$$
$$\implies 15(64 - 9d^2) = 7(64 - d^2) \tag{1.2.24.7}$$
$$\implies 960 - 135d^2 = 448 - 7d^2 \tag{1.2.24.8}$$
$$\implies 135d^2 - 7d^2 = 960 - 448 \tag{1.2.24.9}$$
$$\implies 128d^2 = 512 \tag{1.2.24.10}$$
$$\implies d^2 = \frac{512}{128} = 4 \tag{1.2.24.11}$$
$$\implies d = \pm 2 \tag{1.2.24.12}$$

Taking $d = 2$:

$$a - 3d = 8 - 3(2) = 2 \tag{1.2.24.13}$$
$$a - d = 8 - 2 = 6 \tag{1.2.24.14}$$
$$a + d = 8 + 2 = 10 \tag{1.2.24.15}$$
$$a + 3d = 8 + 3(2) = 14 \tag{1.2.24.16}$$

(Taking $d = -2$ gives the same terms in reverse order: $14, 10, 6, 2$).
Thus, the four numbers are $2, 6, 10$, and 14.

1.2.25 (a) In an equilateral ΔABC, D is a point on side BC such that $BD = \frac{1}{3}BC$. Prove that $9(AD)^2 = 7(AB)^2$.

OR

(b) Prove that, in a right triangle, the square on the hypotenuse is equal to the sum of the squares on the other two sides.

(a) Solution: Let ΔABC be an equilateral triangle with side length $AB = BC = CA = a$. Draw $AE \perp BC$. In an equilateral triangle, the altitude bisects the base, so:

$$BE = EC = \frac{a}{2} \tag{1.2.25.1}$$

Given $BD = \frac{1}{3}BC = \frac{a}{3}$.
Now, calculate DE:

$$DE = BE - BD = \frac{a}{2} - \frac{a}{3} = \frac{a}{6} \tag{1.2.25.2}$$

In right ΔAEB, by Pythagoras theorem, from (1.1.12.1):

$$AE^2 = AB^2 - BE^2 = a^2 - \left(\frac{a}{2}\right)^2 = a^2 - \frac{a^2}{4} = \frac{3a^2}{4} \tag{1.2.25.3}$$

In right ΔAED, by Pythagoras theorem, from (1.1.12.1):

$$AD^2 = AE^2 + DE^2 \tag{1.2.25.4}$$
$$AD^2 = \frac{3a^2}{4} + \left(\frac{a}{6}\right)^2 \tag{1.2.25.5}$$
$$AD^2 = \frac{3a^2}{4} + \frac{a^2}{36} \tag{1.2.25.6}$$
$$AD^2 = \frac{27a^2 + a^2}{36} = \frac{28a^2}{36} = \frac{7a^2}{9} \tag{1.2.25.7}$$
$$\implies 9(AD)^2 = 7a^2 \tag{1.2.25.8}$$

Since $a = AB$:

$$9(AD)^2 = 7(AB)^2 \tag{1.2.25.9}$$

Hence proved.

(b) Solution: Consider a right-angled triangle ABC, right-angled at B.

To prove: $AC^2 = AB^2 + BC^2$.
Construction: Draw $BD \perp AC$.
In ΔADB and ΔABC:

$$\angle ADB = \angle ABC = 90^\circ \tag{1.2.25.10}$$

$$\angle A = \angle A \quad \text{(Common angle)} \tag{1.2.25.11}$$

By AA similarity criterion, $\Delta ADB \sim \Delta ABC$. Therefore, ratio of corresponding sides is equal:

$$\frac{AD}{AB} = \frac{AB}{AC} \implies AB^2 = AD \times AC \tag{1.2.25.12}$$

In ΔBDC and ΔABC:

$$\angle BDC = \angle ABC = 90^\circ \tag{1.2.25.13}$$

$$\angle C = \angle C \quad \text{(Common angle)} \tag{1.2.25.14}$$

By AA similarity criterion, $\Delta BDC \sim \Delta ABC$. Therefore:

$$\frac{DC}{BC} = \frac{BC}{AC} \implies BC^2 = DC \times AC \tag{1.2.25.15}$$

Adding (1.2.25.12) and (1.2.25.15):

$$AB^2 + BC^2 = (AD \times AC) + (DC \times AC) \tag{1.2.25.16}$$

$$= AC(AD + DC) \tag{1.2.25.17}$$

Since $AD + DC = AC$:

$$AB^2 + BC^2 = AC \times AC = AC^2 \tag{1.2.25.18}$$

Hence proved.

1.2.26 Draw a triangle ABC with $BC = 6$ cm, $AB = 5$ cm and $\angle ABC = 60^\circ$. Then construct a triangle whose sides are $\frac{3}{4}$ of the corresponding sides of $\triangle ABC$.
Solution: Steps of construction:

(i) Draw $BC = 6$ cm.
(ii) At B, draw ray BY with $\angle CBY = 60^\circ$. With centre B and radius 5 cm, cut BY at A. Join AC.
(iii) Below BC, draw a ray BX making an acute angle $\angle CBX$ with BC.
(iv) Along BX, mark four points B_1, B_2, B_3, B_4 such that $BB_1 = B_1B_2 = B_2B_3 = B_3B_4$.
(v) Join B_4C. Through B_3, draw $B_3C' \parallel B_4C$ meeting BC at C'.
(vi) Through C', draw $C'A' \parallel CA$ meeting BA at A'.

Then $\Delta A'BC'$ is the required triangle.
Justification: Since $B_3C' \parallel B_4C$, by the basic proportionality theorem

$$\frac{BC'}{BC} = \frac{BB_3}{BB_4} = \frac{3}{4} \tag{1.2.26.1}$$

Since $C'A' \parallel CA$, $\Delta A'BC' \sim \Delta ABC$, so

$$\frac{A'B}{AB} = \frac{BC'}{BC} = \frac{A'C'}{AC} = \frac{3}{4} \tag{1.2.26.2}$$

The following code verifies the construction and generates Fig. 1.2.26.1.

codes/1.2.26.py

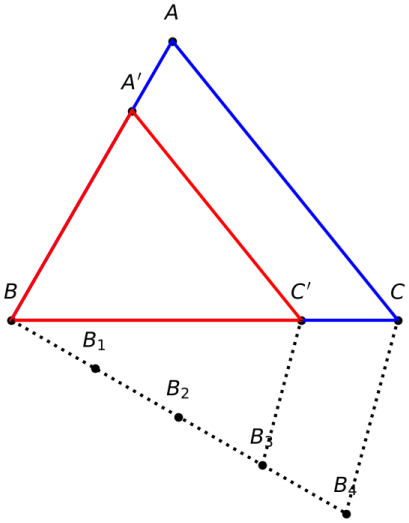


Fig. 1.2.26.1: Construction of $\Delta A'BC'$

1.2.27 Prove that:

$$\frac{\sin A - 2\sin^3 A}{2\cos^3 A - \cos A} = \tan A.$$

Solution: Taking the Left Hand Side (LHS):

$$\text{LHS} = \frac{\sin A - 2\sin^3 A}{2\cos^3 A - \cos A} \tag{1.2.27.1}$$

$$= \frac{\sin A(1 - 2\sin^2 A)}{\cos A(2\cos^2 A - 1)} \tag{1.2.27.2}$$

From (1.1.13.1), $\sin^2 A + \cos^2 A = 1 \implies 1 = \sin^2 A + \cos^2 A$:

$$1 - 2\sin^2 A = (\sin^2 A + \cos^2 A) - 2\sin^2 A \tag{1.2.27.3}$$

$$= \cos^2 A - \sin^2 A \tag{1.2.27.4}$$

$$2\cos^2 A - 1 = 2\cos^2 A - (\sin^2 A + \cos^2 A) \tag{1.2.27.5}$$

$$= \cos^2 A - \sin^2 A \tag{1.2.27.6}$$

Substituting these into the expression:

$$\text{LHS} = \frac{\sin A(\cos^2 A - \sin^2 A)}{\cos A(\cos^2 A - \sin^2 A)} \tag{1.2.27.7}$$

$$= \frac{\sin A}{\cos A} \tag{1.2.27.8}$$

$$= \tan A = \text{RHS} \tag{1.2.27.9}$$

Hence proved.The following code verifies the identity numerically and plots both sides in Fig. 1.2.27.1.

codes/1.2.27.py

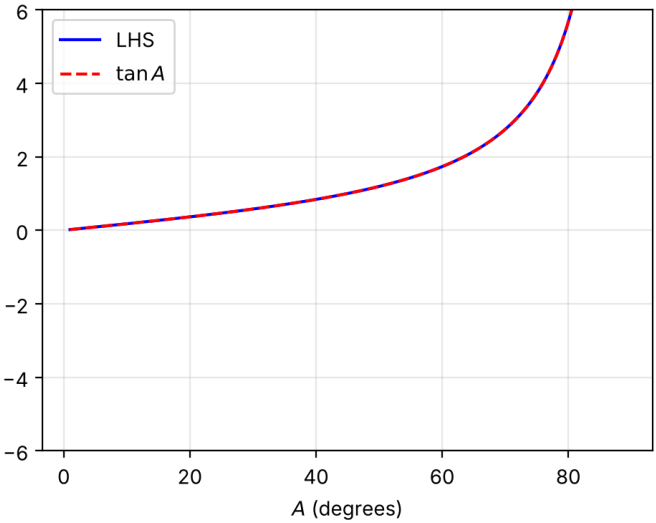


Fig. 1.2.27.1: Plot of LHS and $\tan A$

1.2.28 The diameters of the lower and upper ends of a bucket in the form of a frustum of a cone are 10 cm and 30 cm respectively. If its height is 24 cm, find:
(i) The area of the metal sheet used to make the bucket.
(ii) Why we should avoid the bucket made by ordinary plastic? [Use $\pi = 3.14$]
Solution: Given: Upper radius $R = \frac{30}{2} = 15$ cm, lower radius $r = \frac{10}{2} = 5$ cm, height $h = 24$ cm.

From (1.1.18.1), the slant height l of the frustum is:

$$l = \sqrt{h^2 + (R-r)^2} \tag{1.2.28.1}$$

$$= \sqrt{24^2 + (15-5)^2} \tag{1.2.28.2}$$

$$= \sqrt{576 + 10^2} = \sqrt{576 + 100} = \sqrt{676} = 26 \text{ cm} \tag{1.2.28.3}$$

(i) From (1.1.18.2), area of the metal sheet used = Curved Surface Area + Area of bottom circular base:

$$\text{Area} = \pi(R+r)l + \pi r^2 \tag{1.2.28.4}$$
$$= \pi\left[(15+5)\times 26 + 5^2\right] \tag{1.2.28.5}$$
$$= \pi(20\times 26 + 25) \tag{1.2.28.6}$$
$$= \pi(520+25) = 545\pi \tag{1.2.28.7}$$
$$= 545 \times 3.14 = 1711.3 \text{ cm}^2 \tag{1.2.28.8}$$

Thus, the area of the metal sheet used is 1711.3 cm^2.

(ii) Plastic buckets should be avoided because ordinary plastic is non-biodegradable and causes environmental pollution. Additionally, plastics may leach toxic chemicals when stored with hot water or food items.

1.2.29 As observed from the top of a 100 m high lighthouse from sea level, the angles of depression of two ships are $30°$ and $45°$. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships. [Use $\sqrt{3} = 1.732$]

Solution: Let AB be the lighthouse of height $h = 100$ m, and C, D be the two ships on the same side of the lighthouse such that the angles of depression are $45°$ and $30°$ respectively. Let $BC = x$ m and $CD = y$ m (the distance between the two ships).

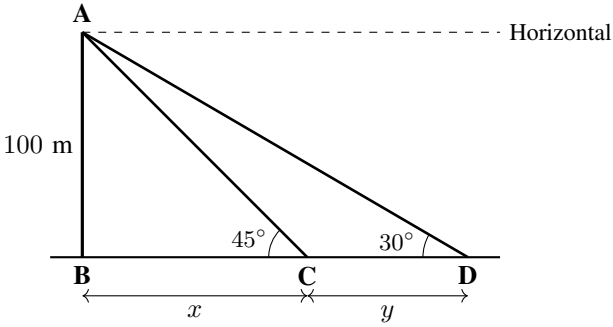


Fig. 1.2.29.1: Lighthouse AB with ships C and D

In right triangle ΔABC, from (1.1.13.4):

$$\tan 45° = \frac{AB}{BC} \tag{1.2.29.1}$$
$$\implies 1 = \frac{100}{x} \tag{1.2.29.2}$$
$$\implies x = 100 \text{ m} \tag{1.2.29.3}$$

In right triangle ΔABD, from (1.1.13.4):

$$\tan 30^\circ = \frac{AB}{BD} = \frac{AB}{x+y} \tag{1.2.29.4}$$

$$\implies \frac{1}{\sqrt{3}} = \frac{100}{100+y} \tag{1.2.29.5}$$

$$\implies 100 + y = 100\sqrt{3} \tag{1.2.29.6}$$

$$\implies y = 100\sqrt{3} - 100 \tag{1.2.29.7}$$

$$\implies y = 100(\sqrt{3} - 1) \tag{1.2.29.8}$$

$$\implies y = 100(1.732 - 1) = 100(0.732) = 73.2 \text{ m} \tag{1.2.29.9}$$

Thus, the distance between the two ships is 73.2 m.

1.2.30 (a) The mean of the following distribution is 18. Find the frequency f of the class 19–21.

Class	11–13	13–15	15–17	17–19	19–21	21–23	23–25
Frequency	3	6	9	13	f	5	4

OR

(b) The following distribution gives the daily income of 50 workers of a factory:

Daily Income (in Rs.)	100–120	120–140	140–160	160–180	180–200
Number of workers	12	14	8	6	10

Convert the distribution above to a less than type cumulative frequency distribution and draw its ogive.

(a) Solution: Constructing the frequency table to calculate the mean:

Class Interval	**Class Mark** (x_i)	**Frequency** (f_i)	$f_i x_i$
11–13	12	3	36
13–15	14	6	84
15–17	16	9	144
17–19	18	13	234
19–21	20	f	$20f$
21–23	22	5	110
23–25	24	4	96
Total		$\sum f_i = 40 + f$	$\sum f_i x_i = 704 + 20f$

Given Mean $(\bar{x}) = 18$. From (1.1.19.1), the mean is:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \tag{1.2.30.1}$$

$$18 = \frac{704 + 20f}{40 + f} \tag{1.2.30.2}$$

$$18(40 + f) = 704 + 20f \tag{1.2.30.3}$$

$$720 + 18f = 704 + 20f \tag{1.2.30.4}$$

$$720 - 704 = 20f - 18f \tag{1.2.30.5}$$

$$16 = 2f \tag{1.2.30.6}$$

$$\implies f = \frac{16}{2} = 8 \tag{1.2.30.7}$$

Thus, the missing frequency f is 8.

(b) Solution: Converting the given data into a "less than type" cumulative frequency distribution:

Daily Income (in Rs.)	**Cumulative Frequency (cf)**
Less than 120	12
Less than 140	26 $(12 + 14)$
Less than 160	34 $(26 + 8)$
Less than 180	40 $(34 + 6)$
Less than 200	50 $(40 + 10)$

To plot the "less than type" ogive, plot the points $(100, 0)$ (the lower limit of the first class, where the cumulative frequency is 0), $(120, 12)$, $(140, 26)$, $(160, 34)$, $(180, 40)$, and $(200, 50)$ on a graph with upper limits on the x-axis and cumulative frequencies on the y-axis, and join them with a smooth freehand curve. The following code verifies part (a) and plots the ogive in Fig. 1.2.30.1.

codes/1.2.30.py

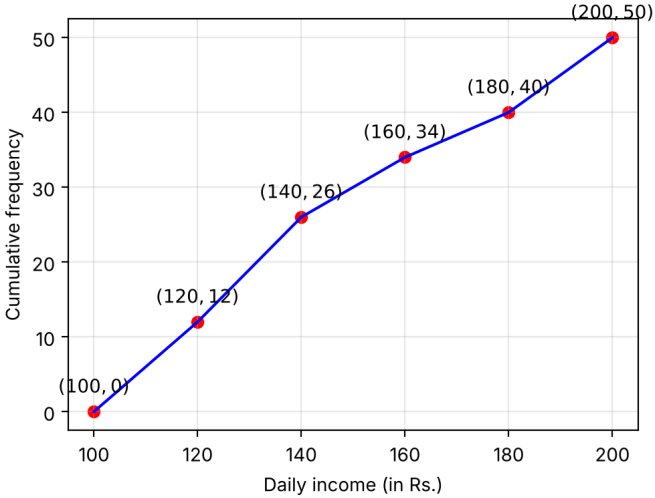


Fig. 1.2.30.1: Less than type ogive for daily income distribution

2. CBSE CLASS-12 MATH 2018

A. Formulae

2.1.1 Determinant properties for square matrices $\mathbf{A}, \mathbf{B}$ of order n and a scalar k:

$$\begin{aligned} \left|\mathbf{AB}\right| &= \left|\mathbf{A}\right|\left|\mathbf{B}\right| \\ \left|\mathbf{A}^\top\right| &= \left|\mathbf{A}\right| \\ \left|k\mathbf{A}\right| &= k^n\left|\mathbf{A}\right| \end{aligned} \tag{2.1.1.1}$$

Solution: If $\mathbf{A}$ is invertible, it is a product of elementary matrices,

$$\mathbf{A} = \mathbf{E}_1\mathbf{E}_2\cdots\mathbf{E}_m \tag{2.1.1.2}$$

and for a single elementary matrix $\left|\mathbf{EB}\right| = \left|\mathbf{E}\right|\left|\mathbf{B}\right|$. Repeating this m times,

$$\left|\mathbf{AB}\right| = \left|\mathbf{E}_1\right|\cdots\left|\mathbf{E}_m\right|\left|\mathbf{B}\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\right| \tag{2.1.1.3}$$

For the transpose, $\mathbf{A}^\top = \mathbf{E}_m^\top\cdots\mathbf{E}_1^\top$ and $\left|\mathbf{E}^\top\right| = \left|\mathbf{E}\right|$ for any elementary matrix, so $\left|\mathbf{A}^\top\right| = \left|\mathbf{A}\right|$. If $\mathbf{A}$ is singular, so is $\mathbf{A}^\top$ and both determinants are 0. Finally, $k\mathbf{A} = (k\mathbf{I})\,\mathbf{A}$ with $\left|k\mathbf{I}\right| = k^n$, yielding (2.1.1.1).

2.1.2 Inverse trigonometric identities:

$$\cos^{-1}(\cos t) = t, \quad t \in [0,\pi] \tag{2.1.2.1}$$

$$\tan^{-1}a + \tan^{-1}b = \tan^{-1}\frac{a+b}{1-ab}, \quad ab < 1 \tag{2.1.2.2}$$

$$\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2} \tag{2.1.2.3}$$

$$\sin^{-1}x = \tan^{-1}\frac{x}{\sqrt{1-x^2}}, \quad |x| < 1 \tag{2.1.2.4}$$

Solution: (2.1.2.1) follows from the definition of the principal value, $\cos^{-1} : [-1,1] \to [0,\pi]$.

For (2.1.2.2), let $\alpha = \tan^{-1}a$ and $\beta = \tan^{-1}b$, both in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$. Then

$$\tan(\alpha+\beta) = \frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta} = \frac{a+b}{1-ab} \tag{2.1.2.5}$$

Also $\cos(\alpha+\beta) = \cos\alpha\cos\beta\,(1-ab) > 0$ for $ab < 1$, and $|\alpha+\beta| < \pi$, so $\alpha+\beta \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, the principal range of $\tan^{-1}$.

For (2.1.2.3), let $\theta = \tan^{-1}x$. Using $\cot(90^\circ-\theta) = \tan\theta = x$ with $\frac{\pi}{2}-\theta \in (0,\pi)$, the principal range of $\cot^{-1}$,

$$\cot^{-1}x = \frac{\pi}{2} - \theta \tag{2.1.2.6}$$

For (2.1.2.4), let $\theta = \sin^{-1}x \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$. Then $\cos\theta = \sqrt{1-x^2} > 0$ and

$$\tan\theta = \frac{x}{\sqrt{1-x^2}} \implies \theta = \tan^{-1}\frac{x}{\sqrt{1-x^2}} \tag{2.1.2.7}$$

2.1.3 Angle θ between a line with direction vector $\mathbf{b}$ and a plane with normal vector $\mathbf{n}$:

$$\sin\theta = \frac{\left|\mathbf{b}^\top \mathbf{n}\right|}{\|\mathbf{b}\|\,\|\mathbf{n}\|} \tag{2.1.3.1}$$

Solution: The normal is perpendicular to the plane, so if ϕ is the angle between $\mathbf{b}$ and $\mathbf{n}$, then $\theta = 90° - \phi$ (or its supplement). From (2.1.4.1),

$$\sin\theta = |\cos\phi| = \frac{\left|\mathbf{b}^\top \mathbf{n}\right|}{\|\mathbf{b}\|\,\|\mathbf{n}\|} \tag{2.1.3.2}$$

2.1.4 For vectors $\mathbf{a}$, $\mathbf{b}$ with angle θ between them:

$$\mathbf{a}^\top \mathbf{b} = \|\mathbf{a}\|\,\|\mathbf{b}\|\cos\theta \tag{2.1.4.1}$$

$$\|\mathbf{a}\times\mathbf{b}\| = \|\mathbf{a}\|\,\|\mathbf{b}\|\sin\theta \tag{2.1.4.2}$$

Solution: The third side of the triangle formed by $\mathbf{a}$ and $\mathbf{b}$ is $\mathbf{a}-\mathbf{b}$. From the cosine rule and by expanding the norm,

$$\|\mathbf{a}-\mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\,\|\mathbf{a}\|\,\|\mathbf{b}\|\cos\theta \tag{2.1.4.3}$$

$$\|\mathbf{a}-\mathbf{b}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\mathbf{a}^\top\mathbf{b} \tag{2.1.4.4}$$

yielding (2.1.4.1). Using Lagrange's identity,

$$\|\mathbf{a}\times\mathbf{b}\|^2 = \|\mathbf{a}\|^2\,\|\mathbf{b}\|^2 - \left(\mathbf{a}^\top\mathbf{b}\right)^2 = \|\mathbf{a}\|^2\,\|\mathbf{b}\|^2\left(1-\cos^2\theta\right) \tag{2.1.4.5}$$

which yields (2.1.4.2) since $\sin\theta \geq 0$ for $\theta \in [0,\pi]$.

2.1.5 A binary operation $*$ on a set S requires $a*b \in S$ for all $a,b\in S$ (closure). It is commutative if

$$a*b = b*a \quad \forall\, a,b\in S \tag{2.1.5.1}$$

Solution: For a finite set $S=\{a_1,\ldots,a_m\}$, form the operation table

$$\mathbf{T} = [T_{ij}], \quad T_{ij} = a_i * a_j \tag{2.1.5.2}$$

The operation is closed if every entry of $\mathbf{T}$ lies in S. It is commutative if

$$T_{ij} = T_{ji} \quad \forall\, i,j \iff \mathbf{T} = \mathbf{T}^\top \tag{2.1.5.3}$$

which is (2.1.5.1).

2.1.6 Homogeneous differential equation $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$: substitute $y = vx$,

$$\frac{dy}{dx} = v + x\frac{dv}{dx} \tag{2.1.6.1}$$

Solution: Differentiating $y=vx$ using the product rule yields (2.1.6.1). Substituting in the equation, the variables separate,

$$x\frac{dv}{dx} = f(v) - v \implies \int \frac{dv}{f(v)-v} = \ln|x| + C \tag{2.1.6.2}$$

Finally v is replaced by $\frac{y}{x}$.

2.1.7 Differential equation of a family of curves with n arbitrary constants $c_1, \ldots, c_n$:

$$F(x, y, c_1, \ldots, c_n) = 0 \implies G\left(x, y, y', \ldots, y^{(n)}\right) = 0 \tag{2.1.7.1}$$

Solution: Differentiating $F = 0$ n times gives $n+1$ equations in the n constants. Eliminating $c_1, \ldots, c_n$ from them leaves an equation of order n. For example, for

$$y = c_1 \cos(\alpha x) + c_2 \sin(\alpha x) \tag{2.1.7.2}$$

differentiating twice,

$$y'' = -\alpha^2 (c_1 \cos(\alpha x) + c_2 \sin(\alpha x)) \implies y'' = -\alpha^2 y \tag{2.1.7.3}$$

which has no arbitrary constants left.

2.1.8 Points A, B, C are collinear if $\mathbf{C} - \mathbf{B} = k(\mathbf{B} - \mathbf{A})$. Points A, B, C, D are coplanar if

$$\operatorname{rank}\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix} = 1 \tag{2.1.8.1}$$

$$\det\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A} & \mathbf{D}-\mathbf{A}\end{pmatrix} = 0 \tag{2.1.8.2}$$

Solution: If $\mathbf{C} - \mathbf{B} = k(\mathbf{B} - \mathbf{A})$, then

$$\mathbf{C} - \mathbf{A} = (\mathbf{C} - \mathbf{B}) + (\mathbf{B} - \mathbf{A}) = (k+1)(\mathbf{B} - \mathbf{A}) \tag{2.1.8.3}$$

so the two columns are linearly dependent, yielding (2.1.8.1). Similarly, if the points are coplanar, then

$$\mathbf{D} - \mathbf{A} = \alpha(\mathbf{B} - \mathbf{A}) + \beta(\mathbf{C} - \mathbf{A}) \tag{2.1.8.4}$$

so the third column is a combination of the first two. Subtracting this combination from the third column leaves a zero column, so the determinant vanishes, yielding (2.1.8.2). Geometrically, the determinant is the volume of the parallelepiped on the three vectors.

2.1.9 Standard integrals:

$$\int e^x [f(x) + f'(x)]\, dx = e^x f(x) + C \tag{2.1.9.1}$$

$$\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1}\frac{u}{a} + C \tag{2.1.9.2}$$

$$\int \sqrt{a^2 - u^2}\, du = \frac{u}{2}\sqrt{a^2 - u^2} + \frac{a^2}{2}\sin^{-1}\frac{u}{a} + C \tag{2.1.9.3}$$

$$2\cos A \cos B = \cos(A + B) + \cos(A - B) \tag{2.1.9.4}$$

Solution: Differentiating $e^x f(x)$ using the product rule gives $e^x [f + f']$, yielding (2.1.9.1).

For (2.1.9.2) and (2.1.9.3), let $u = a\sin\theta$, so $du = a\cos\theta\, d\theta$ and $\sqrt{a^2 - u^2} =$

$a\cos\theta$,

$$\int \frac{du}{\sqrt{a^2-u^2}} = \int d\theta = \theta = \sin^{-1}\frac{u}{a} \tag{2.1.9.5}$$

$$\int \sqrt{a^2-u^2}\,du = a^2\int \cos^2\theta\,d\theta = \frac{a^2}{2}\left(\theta+\sin\theta\cos\theta\right) \tag{2.1.9.6}$$

and $\sin\theta\cos\theta = \frac{u}{a}\cdot\frac{\sqrt{a^2-u^2}}{a}$.

For (2.1.9.4), let $\mathbf{R}(\theta) = \begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$ be the rotation matrix. Then $\mathbf{R}(A)\,\mathbf{R}(B) = \mathbf{R}(A+B)$, and comparing the first entries,

$$\cos(A+B) = \cos A\cos B - \sin A\sin B \tag{2.1.9.7}$$

Replacing B by $-B$ gives $\cos(A-B) = \cos A\cos B + \sin A\sin B$. Adding the two yields (2.1.9.4).

2.1.10 The plane through a point $\mathbf{P}_0$ with normal $\mathbf{n}$ is

$$\mathbf{n}^\top(\mathbf{x}-\mathbf{P}_0) = 0 \tag{2.1.10.1}$$

A plane perpendicular to two planes with normals $\mathbf{n}_1, \mathbf{n}_2$ has normal

$$\mathbf{n} = \mathbf{n}_1\times\mathbf{n}_2 \tag{2.1.10.2}$$

Solution: A point $\mathbf{x}$ lies on the plane if $\mathbf{x}-\mathbf{P}_0$ is perpendicular to $\mathbf{n}$, yielding (2.1.10.1). Two planes are perpendicular if their normals are perpendicular, so

$$\mathbf{n}^\top\mathbf{n}_1 = 0, \quad \mathbf{n}^\top\mathbf{n}_2 = 0 \iff \begin{pmatrix}\mathbf{n}_1^\top\\ \mathbf{n}_2^\top\end{pmatrix}\mathbf{n} = \mathbf{0} \tag{2.1.10.3}$$

The vector $\mathbf{n}_1\times\mathbf{n}_2$ is perpendicular to both $\mathbf{n}_1$ and $\mathbf{n}_2$. For non-parallel $\mathbf{n}_1, \mathbf{n}_2$ the solution space of this system is one-dimensional, yielding (2.1.10.2).

2.1.11 Inverse of a matrix and elementary row operations:

$$\mathbf{A}^{-1} = \frac{1}{\left|\mathbf{A}\right|}\operatorname{adj}(\mathbf{A}), \quad \left|\mathbf{A}\right| \neq 0 \tag{2.1.11.1}$$

$$[\mathbf{A}\,|\,\mathbf{I}] \xrightarrow{\text{row operations}} [\mathbf{I}\,|\,\mathbf{A}^{-1}] \tag{2.1.11.2}$$

A system $\mathbf{A}\mathbf{x} = \mathbf{b}$ with $\left|\mathbf{A}\right| \neq 0$ has the unique solution

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} \tag{2.1.11.3}$$

Solution: Let C_{jk} be the cofactor of a_{jk}, so that $\operatorname{adj}(\mathbf{A})_{kj} = C_{jk}$. Expanding the determinant along row j, and noting that a determinant with two equal rows is zero,

$$\left(\mathbf{A}\operatorname{adj}(\mathbf{A})\right)_{ij} = \sum_k a_{ik}C_{jk} = \left|\mathbf{A}\right|\delta_{ij} \implies \mathbf{A}\operatorname{adj}(\mathbf{A}) = \left|\mathbf{A}\right|\mathbf{I} \tag{2.1.11.4}$$

yielding (2.1.11.1). Each row operation is a left multiplication by an elementary

matrix, so a sequence of them is $\mathbf{E}$ and

$$\mathbf{E}\left[\mathbf{A}\,|\,\mathbf{I}\right]=\left[\mathbf{E}\mathbf{A}\,|\,\mathbf{E}\right] \tag{2.1.11.5}$$

If $\mathbf{E}\mathbf{A}=\mathbf{I}$, then $\mathbf{E}=\mathbf{A}^{-1}$, yielding (2.1.11.2). Multiplying $\mathbf{A}\mathbf{x}=\mathbf{b}$ by $\mathbf{A}^{-1}$ gives (2.1.11.3).

2.1.12 A relation R on a set S is an equivalence relation if it is reflexive, symmetric and transitive:

$$aRa,\quad aRb\implies bRa,\quad aRb \text{ and } bRc\implies aRc \tag{2.1.12.1}$$

A function $f:X\to Y$ is invertible if it is one-one and onto.

Solution: For a finite set, let $\mathbf{M}$ be the relation matrix with $M_{ij}=1$ if a_iRa_j and 0 otherwise. Then R is reflexive if the diagonal of $\mathbf{M}$ is all ones, and symmetric if $\mathbf{M}=\mathbf{M}^{\top}$. For example, on the integers let aRb if 3 divides $a-b$. Then $a-a=0$, and for $a-b=3m$ and $b-c=3n$,

$$b-a=-3m,\quad a-c=(a-b)+(b-c)=3\left(m+n\right) \tag{2.1.12.2}$$

so R is an equivalence relation. For a function, f one-one means $f\left(x_1\right)=f\left(x_2\right)\implies x_1=x_2$, so each y has at most one preimage. Onto means each y has at least one. Hence $f^{-1}\left(y\right)$ is well defined. For finite sets, the matrix of f is then a permutation matrix $\mathbf{P}$ with

$$\mathbf{P}^{-1}=\mathbf{P}^{\top} \tag{2.1.12.3}$$

2.1.13 Logarithmic differentiation of $y=u\left(x\right)^{v(x)}$, $u>0$:

$$\ln y=v\ln u\implies\frac{1}{y}\frac{dy}{dx}=v'\ln u+\frac{v\,u'}{u} \tag{2.1.13.1}$$

Solution: Taking the natural logarithm of $y=u^v$ gives $\ln y=v\ln u$. Differentiating both sides with respect to x, using the chain rule on the left and the product rule on the right,

$$\frac{1}{y}\frac{dy}{dx}=v'\ln u+v\cdot\frac{u'}{u} \tag{2.1.13.2}$$

yielding (2.1.13.1). Hence $\frac{dy}{dx}=u^v\left(v'\ln u+\frac{vu'}{u}\right)$.

2.1.14 If $f'\left(x\right)<0$ on an interval, then f is strictly decreasing on that interval. At a critical point $f'\left(x_0\right)=0$, if $f''\left(x_0\right)<0$ then f has a maximum.

$$f'\left(x\right)<0\implies f\left(x_2\right)<f\left(x_1\right),\quad x_1<x_2 \tag{2.1.14.1}$$

Solution: By the mean value theorem, for some $\xi\in(x_1,x_2)$,

$$f\left(x_2\right)-f\left(x_1\right)=f'\left(\xi\right)\left(x_2-x_1\right)<0 \tag{2.1.14.2}$$

yielding (2.1.14.1). At a critical point, the second order Taylor expansion gives

$$f\left(x_0+h\right)-f\left(x_0\right)\approx\frac{1}{2}f''\left(x_0\right)h^2<0 \tag{2.1.14.3}$$

for small $h \neq 0$, so $f(x_0)$ is a maximum.

2.1.15 Area between two curves and area of a triangle with vertices $\mathbf{A}, \mathbf{B}, \mathbf{C}$:

$$\text{Area} = \int_a^b [y_1(x) - y_2(x)]\, dx \tag{2.1.15.1}$$

$$\text{Area} = \int_c^d [x_1(y) - x_2(y)]\, dy \tag{2.1.15.2}$$

$$\text{Area}(\Delta ABC) = \frac{1}{2} \left| \det \begin{pmatrix} \mathbf{B} - \mathbf{A} & \mathbf{C} - \mathbf{A} \end{pmatrix} \right| \tag{2.1.15.3}$$

Solution: Divide $[a, b]$ into strips of width Δx. For $y_1 \geq y_2$, each strip is approximately a rectangle of height $y_1(x_i) - y_2(x_i)$, so

$$\text{Area} = \lim_{\Delta x \to 0} \sum_i [y_1(x_i) - y_2(x_i)]\, \Delta x \tag{2.1.15.4}$$

which is (2.1.15.1). Interchanging the roles of x and y gives (2.1.15.2). For the triangle, let $\mathbf{a} = \mathbf{B} - \mathbf{A}$, $\mathbf{b} = \mathbf{C} - \mathbf{A}$ and θ the angle at A. Writing $\mathbf{a} = \|\mathbf{a}\| \begin{pmatrix} \cos\phi \\ \sin\phi \end{pmatrix}$ and $\mathbf{b} = \|\mathbf{b}\| \begin{pmatrix} \cos(\phi + \theta) \\ \sin(\phi + \theta) \end{pmatrix}$,

$$\det \begin{pmatrix} \mathbf{a} & \mathbf{b} \end{pmatrix} = \|\mathbf{a}\| \, \|\mathbf{b}\| \, [\cos\phi \sin(\phi + \theta) - \sin\phi \cos(\phi + \theta)] = \|\mathbf{a}\| \, \|\mathbf{b}\| \sin\theta \tag{2.1.15.5}$$

and $\text{Area} = \frac{1}{2} \|\mathbf{a}\| \, \|\mathbf{b}\| \sin\theta$, yielding (2.1.15.3).

2.1.16 Distance between the parallel lines $\mathbf{x} = \mathbf{a}_1 + \lambda \mathbf{b}$ and $\mathbf{x} = \mathbf{a}_2 + \mu \mathbf{b}$:

$$d = \frac{\|(\mathbf{a}_2 - \mathbf{a}_1) \times \mathbf{b}\|}{\|\mathbf{b}\|} \tag{2.1.16.1}$$

Foot of the perpendicular $\mathbf{Q}$ from $\mathbf{P}$ to the plane $\mathbf{n}^\top \mathbf{x} = c$, and the image $\mathbf{P}'$ of $\mathbf{P}$:

$$\mathbf{Q} = \mathbf{P} + t\mathbf{n}, \quad t = \frac{c - \mathbf{n}^\top \mathbf{P}}{\|\mathbf{n}\|^2}, \quad \mathbf{P}' = \mathbf{P} + 2t\mathbf{n} \tag{2.1.16.2}$$

Solution: The distance is that from $\mathbf{a}_2$ to the first line. Let $\mathbf{w} = \mathbf{a}_2 - \mathbf{a}_1$. The foot of the perpendicular is at λ where

$$\mathbf{b}^\top (\mathbf{w} - \lambda \mathbf{b}) = 0 \implies \lambda = \frac{\mathbf{b}^\top \mathbf{w}}{\|\mathbf{b}\|^2} \tag{2.1.16.3}$$

and the distance is the length of the remaining component,

$$d^2 = \|\mathbf{w} - \lambda \mathbf{b}\|^2 = \|\mathbf{w}\|^2 - \frac{(\mathbf{b}^\top \mathbf{w})^2}{\|\mathbf{b}\|^2} = \frac{\|\mathbf{w} \times \mathbf{b}\|^2}{\|\mathbf{b}\|^2} \tag{2.1.16.4}$$

using Lagrange's identity, yielding (2.1.16.1). For the plane, $\mathbf{Q}$ lies along the normal

from $\mathbf{P}$ and on the plane, so

$$\mathbf{n}^\top(\mathbf{P}+t\mathbf{n})=c \implies t=\frac{c-\mathbf{n}^\top\mathbf{P}}{\|\mathbf{n}\|^2} \tag{2.1.16.5}$$

The foot $\mathbf{Q}$ is the midpoint of $\mathbf{P}$ and $\mathbf{P}'$, so $\mathbf{P}'=2\mathbf{Q}-\mathbf{P}=\mathbf{P}+2t\mathbf{n}$, yielding (2.1.16.2).

2.1.17 Linear programming: the optimum of a linear objective over a bounded convex feasible region occurs at a corner point.

$$\max Z=\mathbf{c}^\top\mathbf{x} \quad \text{s.t.} \quad \mathbf{A}\mathbf{x}\le\mathbf{b},\ \mathbf{x}\ge\mathbf{0} \tag{2.1.17.1}$$

Solution: Let $\mathbf{v}_1,\ldots,\mathbf{v}_m$ be the corner points. Any feasible point is a convex combination of them,

$$\mathbf{x}=\sum_{i=1}^{m}\lambda_i\mathbf{v}_i, \quad \lambda_i\ge 0, \quad \sum_{i=1}^{m}\lambda_i=1 \tag{2.1.17.2}$$

Then

$$Z(\mathbf{x})=\mathbf{c}^\top\mathbf{x}=\sum_{i=1}^{m}\lambda_i\mathbf{c}^\top\mathbf{v}_i\le\max_i\ \mathbf{c}^\top\mathbf{v}_i \tag{2.1.17.3}$$

so the maximum is attained at a corner point.

2.1.18 Cylinder inscribed in a cone of base radius R and height H: if the cylinder has radius r and height h, then

$$h=\frac{H(R-r)}{R}, \quad \text{CSA}=2\pi rh \tag{2.1.18.1}$$

Solution: The top rim of the cylinder lies on the cone at height h, where the cone has radius r. By similar triangles,

$$\frac{H-h}{H}=\frac{r}{R} \implies h=\frac{H(R-r)}{R} \tag{2.1.18.2}$$

yielding (2.1.18.1). To maximize the CSA,

$$\text{CSA}(r)=\frac{2\pi H}{R}\left(Rr-r^2\right) \tag{2.1.18.3}$$

$$\frac{d\,\text{CSA}}{dr}=\frac{2\pi H}{R}(R-2r)=0 \implies r=\frac{R}{2} \tag{2.1.18.4}$$

Since $\frac{d^2\,\text{CSA}}{dr^2}=-\frac{4\pi H}{R}<0$, this is a maximum, with $h=\frac{H}{2}$ and $\text{CSA}_{\max}=\frac{\pi RH}{2}$.

2.1.19 Gaussian elimination for $A\mathbf{x}=\mathbf{b}$, where A is an $n\times n$ matrix. Forward elimination reduces $[A\,|\,\mathbf{b}]$ to the upper triangular form $[U\,|\,\mathbf{c}]$ using

$$R_i\to R_i-m_{ik}R_k, \quad m_{ik}=\frac{a_{ik}}{a_{kk}}, \quad i>k \tag{2.1.19.1}$$

and back-substitution gives

$$x_i = \frac{1}{u_{ii}}\left(c_i - \sum_{j>i} u_{ij}x_j\right), \quad i = n, n-1, \ldots, 1 \tag{2.1.19.2}$$

The number of operations, and so the running time, grows as

$$T(n) \approx \kappa\, n^3 \tag{2.1.19.3}$$

2.1.20 Probability mass function (PMF) of a discrete random variable X, and of a pair (X, Y):

$$p_X(k) = \Pr(X = k), \quad \sum_k p_X(k) = 1, \quad \Pr(X \in E) = \sum_{k\in E} p_X(k) \tag{2.1.20.1}$$

$$p_{XY}(i, j) = p_X(i)\, p_Y(j) \quad \text{when } X \text{ and } Y \text{ are independent} \tag{2.1.20.2}$$

2.1.21 Laplace transform, with $F(s) = \mathcal{L}\{f\}(s)$, and the integration property:

$$F(s) = \int_0^\infty f(x)e^{-sx}\,dx, \quad \mathcal{L}\{x^n\} = \frac{n!}{s^{n+1}}, \quad \mathcal{L}\{\cos\omega x\} = \frac{s}{s^2+\omega^2} \tag{2.1.21.1}$$

$$\mathcal{L}\left\{\int_0^x f(\tau)\,d\tau\right\} = \frac{F(s)}{s} \tag{2.1.21.2}$$

So $\int f(x)\,dx = \mathcal{L}^{-1}\{F(s)/s\} + C$, where C is the value of the integral at $x = 0$.

2.1.22 Laplace transform of derivatives and of shifted ramps, with u the unit step:

$$\mathcal{L}\{y'\} = sY - y(0), \quad \mathcal{L}\{xf(x)\} = -F'(s) \tag{2.1.22.1}$$

$$\mathcal{L}\{(x-a)^n u(x-a)\} = \frac{n!\,e^{-as}}{s^{n+1}}, \quad |x-a| = (a-x) + 2(x-a)u(x-a) \tag{2.1.22.2}$$

B. Problems

2.2.1 If A is a square matrix of order 2 and $|A| = 4$, then find the value of $|2 \cdot AA'|$, where A' is the transpose of matrix A.

Solution: From (2.1.1.1) with $n = 2$ and $k = 2$:

$$|2 \cdot AA'| = 2^2|AA'| \tag{2.2.1.1}$$

$$= 4|A||A'| = 4|A|^2 \tag{2.2.1.2}$$

$$= 4(4)^2 = 64 \tag{2.2.1.3}$$

2.2.2 (a) Find the angle between the line $\mathbf{r} = (2\hat{i} - \hat{j} + 3\hat{k}) + \lambda(3\hat{i} - \hat{j} + 2\hat{k})$ and the plane $\mathbf{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 3$.

OR

(b) Find the co-ordinates of the point, where the line $\dfrac{x+2}{1} = \dfrac{y-5}{3} = \dfrac{z+1}{5}$ cuts the yz-plane.

(a) Solution: The direction vector of the line and the normal of the plane are

$$\mathbf{b} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \tag{2.2.2.1}$$

From (2.1.3.1):

$$\sin\theta = \frac{|(3)(1) + (-1)(1) + (2)(1)|}{\sqrt{3^2 + (-1)^2 + 2^2}\sqrt{1^2 + 1^2 + 1^2}} \tag{2.2.2.2}$$

$$= \frac{4}{\sqrt{14}\sqrt{3}} = \frac{4}{\sqrt{42}} \tag{2.2.2.3}$$

$$\implies \theta = \sin^{-1}\left(\frac{4}{\sqrt{42}}\right) \approx 38.11^\circ \tag{2.2.2.4}$$

(b) Solution: On the yz-plane $x = 0$. Let the common ratio be λ:

$$\frac{0+2}{1} = \lambda = 2 \implies y - 5 = 3(2) \implies y = 11 \tag{2.2.2.5}$$

$$\implies z + 1 = 5(2) \implies z = 9 \tag{2.2.2.6}$$

Thus, the point is $(0, 11, 9)$. The following code verifies both parts and generates Fig. 2.2.2.1.

codes/2.2.2.py

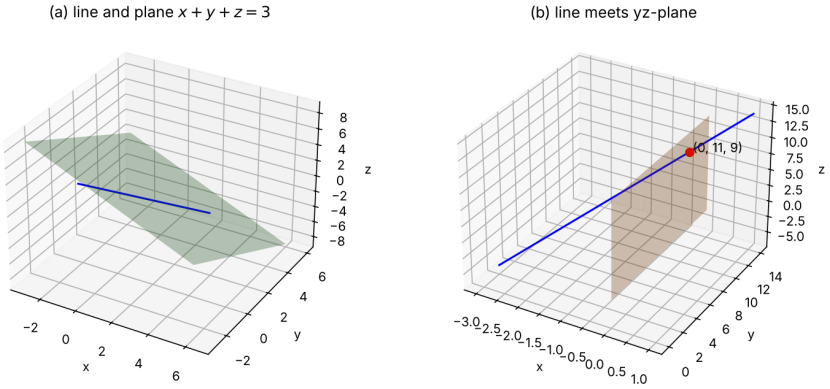


Fig. 2.2.2.1: (a) Line and plane $x + y + z = 3$; (b) line meeting the yz-plane at $(0, 11, 9)$

2.2.3 Find the differential equation representing the family of curves $y = -A\cos 3x + B\sin 3x$.

Solution: The family has two arbitrary constants A and B, so by 2.1.7.1 we

differentiate twice:

$$\frac{dy}{dx} = 3A\sin 3x + 3B\cos 3x \tag{2.2.3.1}$$

$$\frac{d^2y}{dx^2} = 9A\cos 3x - 9B\sin 3x = -9\left(-A\cos 3x + B\sin 3x\right) = -9y \tag{2.2.3.2}$$

Hence the differential equation is

$$\frac{d^2y}{dx^2} + 9y = 0 \tag{2.2.3.3}$$

The following code verifies this symbolically and plots members of the family in Fig. 2.2.3.1.

codes/2.2.3.py

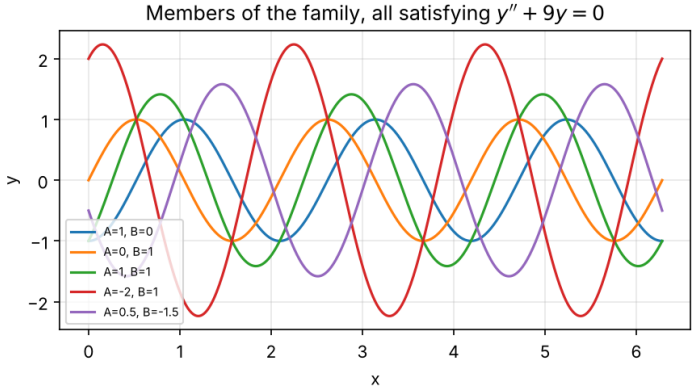


Fig. 2.2.3.1: Members of the family $y = -A\cos 3x + B\sin 3x$

2.2.4 Find the differential of the function $\cos^{-1}(\sin 2x)$ w.r.t. x.

Solution: Since $\sin 2x = \cos\left(\frac{\pi}{2} - 2x\right)$, from (2.1.2.1), for $\frac{\pi}{2} - 2x \in [0, \pi]$, i.e. $-\frac{\pi}{4} \le x \le \frac{\pi}{4}$:

$$y = \cos^{-1}\left(\cos\left(\frac{\pi}{2} - 2x\right)\right) = \frac{\pi}{2} - 2x \tag{2.2.4.1}$$

$$\implies \frac{dy}{dx} = -2 \tag{2.2.4.2}$$

The following code verifies the derivative numerically and plots Fig. 2.2.4.1.

codes/2.2.4.py

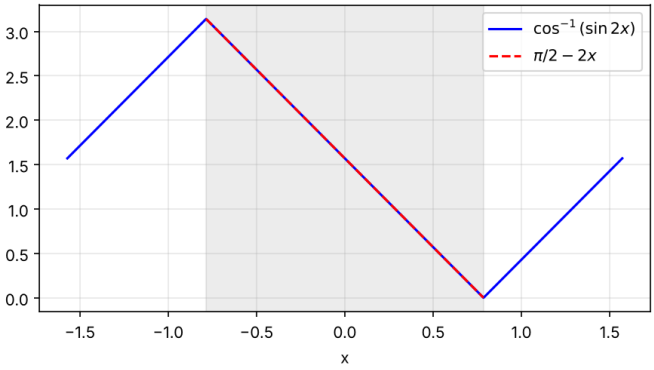


Fig. 2.2.4.1: Plot of $\cos^{-1}(\sin 2x)$; it is the line $\frac{\pi}{2} - 2x$ on the shaded region

2.2.5 If an operation $*$ on the set of integers $\mathbb{Z}$ is defined by $a * b = 2a^2 + b$, then find (i) whether it is a binary or not, and (ii) if a binary, then is it commutative or not.
Solution: (i) For any $a, b \in \mathbb{Z}$, $2a^2 + b \in \mathbb{Z}$, so $*$ maps $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ and is a binary operation.

(ii) From (2.1.5.1):

$$a * b = 2a^2 + b, \quad b * a = 2b^2 + a \tag{2.2.5.1}$$

For $a = 1$, $b = 2$:

$$1 * 2 = 2(1)^2 + 2 = 4, \quad 2 * 1 = 2(2)^2 + 1 = 9 \implies 1 * 2 \neq 2 * 1 \tag{2.2.5.2}$$

Hence $*$ is a binary operation but it is not commutative.

2.2.6 (a) Four cards are drawn one by one with replacement from a well-shuffled deck of playing cards. Find the probability that at least three cards are of diamonds.

OR

(b) The probability of two students A and B coming to school on time are $\frac{2}{7}$ and $\frac{4}{7}$, respectively. Assuming that the events 'A coming on time' and 'B coming on time' are independent, find the probability of only one of them coming to school on time.

(a) Solution: Let X be the number of diamonds in 4 draws with replacement. Then X is binomial with $n = 4$, $p = \frac{1}{4}$, and its PMF from (2.1.20.1) is

$$p_X(k) = \binom{4}{k} \left(\frac{1}{4}\right)^k \left(\frac{3}{4}\right)^{4-k}, \quad k = 0, 1, \ldots, 4 \tag{2.2.6.1}$$

$$\Pr(X \geq 3) = p_X(3) + p_X(4) = \frac{12}{256} + \frac{1}{256} = \frac{13}{256} \tag{2.2.6.2}$$

(b) Solution: Let $A, B \in \{0, 1\}$ indicate that each student is on time, with $p_A(1) = \frac{2}{7}$

and $p_B(1) = \frac{4}{7}$. By independence, from (2.1.20.2), $p_{AB}(a,b) = p_A(a)p_B(b)$, so

$$p_{AB}(1,0) = \frac{2}{7}\cdot\frac{3}{7} = \frac{6}{49}, \quad p_{AB}(0,1) = \frac{5}{7}\cdot\frac{4}{7} = \frac{20}{49} \tag{2.2.6.3}$$

$$\Pr\,(\text{only one}) = p_{AB}(1,0) + p_{AB}(0,1) = \frac{26}{49} \tag{2.2.6.4}$$

The following code verifies both parts (exact fractions and a simulation) and plots Fig. 2.2.6.1.

codes/2.2.6.py

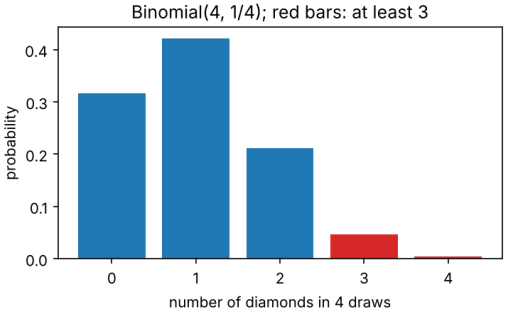


Fig. 2.2.6.1: Distribution of the number of diamonds in four draws

2.2.7 Find the value of $(x-y)$ from the matrix equation

$$2\begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} -3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}.$$

Solution: Moving the constant matrix to the right and dividing by 2,

$$2\begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix} - \begin{bmatrix} -3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 10 & 10 \\ 14 & 12 \end{bmatrix} \tag{2.2.7.1}$$

$$\begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ 7 & 6 \end{bmatrix} \tag{2.2.7.2}$$

Comparing entries, $x = 5$ and $y - 3 = 6 \implies y = 9$. Hence

$$x - y = 5 - 9 = -4 \tag{2.2.7.3}$$

2.2.8 Solve the following differential equation:

$$(y + 3x^2)\frac{dx}{dy} = x$$

Solution: Taking reciprocals, $\frac{dy}{dx} = \frac{y+3x^2}{x}$, that is,

$$xy' - y = 3x^2 \tag{2.2.8.1}$$

The equation is singular at $x = 0$, so we look for solutions with $y(0) = 0$ and let $y'(0) = C$. With $Y(s) = \mathcal{L}\{y\}$, (2.1.22.1) gives $\mathcal{L}\{y'\} = sY$ and so $\mathcal{L}\{xy'\} = -\frac{d}{ds}(sY) = -sY' - Y$. Using $\mathcal{L}\{3x^2\} = \frac{6}{s^3}$ from (2.1.21.1),

$$-sY' - Y - Y = \frac{6}{s^3} \tag{2.2.8.2}$$

$$\implies s^2Y' + 2sY = \left(s^2Y\right)' = -\frac{6}{s^2} \tag{2.2.8.3}$$

Integrating, $s^2Y = \frac{6}{s} + C$, so

$$Y(s) = \frac{6}{s^3} + \frac{C}{s^2} \implies y = 3x^2 + Cx \tag{2.2.8.4}$$

The following code verifies the solution and plots Fig. 2.2.8.1.

codes/2.2.8.py

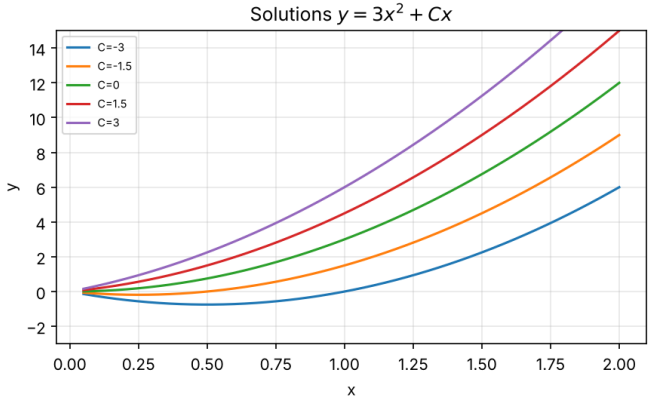


Fig. 2.2.8.1: Solutions $y = 3x^2 + Cx$ for several values of C

2.2.9 (a) Using vectors, prove that the points $(2, -1, 3)$, $(3, -5, 1)$ and $(-1, 11, 9)$ are collinear.

OR

(b) For any two vectors $\mathbf{a}$ and $\mathbf{b}$, prove that $(\mathbf{a} \times \mathbf{b})^2 = \mathbf{a}^2\mathbf{b}^2 - (\mathbf{a} \cdot \mathbf{b})^2$.

(a) Solution: Let $A(2,-1,3)$, $B(3,-5,1)$, $C(-1,11,9)$.

$$\mathbf{AB} = \begin{pmatrix} 3-2 \\ -5+1 \\ 1-3 \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \\ -2 \end{pmatrix} \tag{2.2.9.1}$$

$$\mathbf{BC} = \begin{pmatrix} -1-3 \\ 11+5 \\ 9-1 \end{pmatrix} = \begin{pmatrix} -4 \\ 16 \\ 8 \end{pmatrix} = -4\begin{pmatrix} 1 \\ -4 \\ -2 \end{pmatrix} = -4\,\mathbf{AB} \tag{2.2.9.2}$$

Since $\mathbf{BC} = k\,\mathbf{AB}$ with $k = -4$ and B is common to both, the points A, B, C are collinear.

(b) Solution: Let θ be the angle between $\mathbf{a}$ and $\mathbf{b}$. From (2.1.4.2),

$$\|\mathbf{a}\times\mathbf{b}\|^2 = \|\mathbf{a}\|^2\|\mathbf{b}\|^2\sin^2\theta \tag{2.2.9.3}$$

$$= \|\mathbf{a}\|^2\|\mathbf{b}\|^2\left(1-\cos^2\theta\right) \tag{2.2.9.4}$$

$$= \|\mathbf{a}\|^2\|\mathbf{b}\|^2 - \left(\|\mathbf{a}\|\|\mathbf{b}\|\cos\theta\right)^2 \tag{2.2.9.5}$$

and, from (2.1.4.1), $\|\mathbf{a}\|\|\mathbf{b}\|\cos\theta = \mathbf{a}^\top\mathbf{b}$. Hence

$$\|\mathbf{a}\times\mathbf{b}\|^2 = \|\mathbf{a}\|^2\|\mathbf{b}\|^2 - \left(\mathbf{a}^\top\mathbf{b}\right)^2 \tag{2.2.9.6}$$

The following code verifies both parts and generates Fig. 2.2.9.1.

codes/2.2.9.py

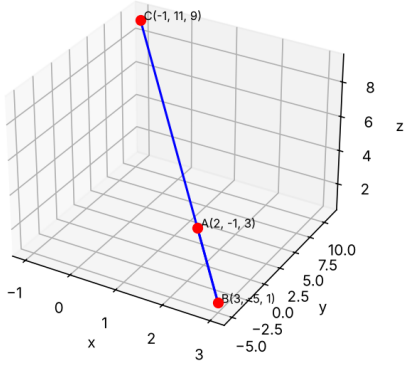


Fig. 2.2.9.1: The collinear points A, B, C

2.2.10 (a) Find: $\displaystyle\int \frac{x-1}{(x-2)(x-3)}\,dx$

OR

(b) Integrate: $\displaystyle\frac{e^x}{\sqrt{5-4e^x-e^{2x}}}$ with respect to x.

(a) Solution: Using partial fractions, $\frac{x-1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$:

$$A = \frac{2-1}{2-3} = -1, \quad B = \frac{3-1}{3-2} = 2 \tag{2.2.10.1}$$

$$\int \frac{x-1}{(x-2)(x-3)}dx = -\int \frac{dx}{x-2} + 2\int \frac{dx}{x-3} \tag{2.2.10.2}$$

$$= -\ln|x-2| + 2\ln|x-3| + C \tag{2.2.10.3}$$

(b) Solution: Let $t = e^x$, so $dt = e^x dx$. Since $5 - 4t - t^2 = 9 - (t+2)^2$, from (2.1.9.2):

$$\int \frac{dt}{\sqrt{5-4t-t^2}} = \int \frac{dt}{\sqrt{9-(t+2)^2}} \tag{2.2.10.4}$$

$$= \sin^{-1}\left(\frac{t+2}{3}\right) + C = \sin^{-1}\left(\frac{e^x+2}{3}\right) + C \tag{2.2.10.5}$$

(The integrand is real only for $e^x < 1$, i.e. $x < 0$.) The following code verifies both antiderivatives by differentiation.

codes/2.2.10.py

2.2.11 Find: $\int e^x \left(\frac{2+\sin 2x}{2\cos^2 x}\right) dx$

Solution: Using $\sin 2x = 2\sin x \cos x$:

$$\frac{2+\sin 2x}{2\cos^2 x} = \frac{1}{\cos^2 x} + \frac{2\sin x\cos x}{2\cos^2 x} = \sec^2 x + \tan x \tag{2.2.11.1}$$

With $f(x) = \tan x$ and $f'(x) = \sec^2 x$, from (2.1.9.1):

$$\int e^x(\tan x + \sec^2 x)\,dx = e^x \tan x + C \tag{2.2.11.2}$$

The following code verifies the result by differentiation.

codes/2.2.11.py

2.2.12 If A and B are independent events with $P(A) = \frac{3}{7}$ and $P(B) = \frac{2}{5}$, then find $P(A' \cap B')$.

Solution: Let $A, B \in \{0,1\}$ be the indicators of the two events, with $p_A(1) = \frac{3}{7}$ and $p_B(1) = \frac{2}{5}$. By independence, from (2.1.20.2), $p_{AB}(a,b) = p_A(a)p_B(b)$. Neither event occurs when $(A,B) = (0,0)$:

$$p_{AB}(0,0) = \frac{4}{7} \cdot \frac{3}{5} = \frac{12}{35} \tag{2.2.12.1}$$

Hence $P(A' \cap B') = \frac{12}{35}$.

2.2.13 Find the equation of the plane passing through the point $(-1,3,2)$ and perpendicular to the planes $x + 2y + 3z = 5$ and $3x + 3y + z = 0$.

Solution: The normals of the given planes are $\mathbf{n}_1 = \begin{pmatrix}1\\2\\3\end{pmatrix}$ and $\mathbf{n}_2 = \begin{pmatrix}3\\3\\1\end{pmatrix}$. From (2.1.10.2),

$$\mathbf{n} = \mathbf{n}_1 \times \mathbf{n}_2 = \begin{pmatrix}2\cdot 1 - 3\cdot 3\\3\cdot 3 - 1\cdot 1\\1\cdot 3 - 2\cdot 3\end{pmatrix} = \begin{pmatrix}-7\\8\\-3\end{pmatrix} \tag{2.2.13.1}$$

The plane through $\mathbf{P}_0 = \begin{pmatrix}-1\\3\\2\end{pmatrix}$ is $\mathbf{n}^\top(\mathbf{x} - \mathbf{P}_0) = 0$:

$$-7(x+1) + 8(y-3) - 3(z-2) = 0 \tag{2.2.13.2}$$
$$\implies -7x + 8y - 3z - 25 = 0 \tag{2.2.13.3}$$
$$\implies 7x - 8y + 3z + 25 = 0 \tag{2.2.13.4}$$

The following code verifies $\mathbf{n} \perp \mathbf{n}_1, \mathbf{n}_2$ and plots Fig. 2.2.13.1.

codes/2.2.13.py

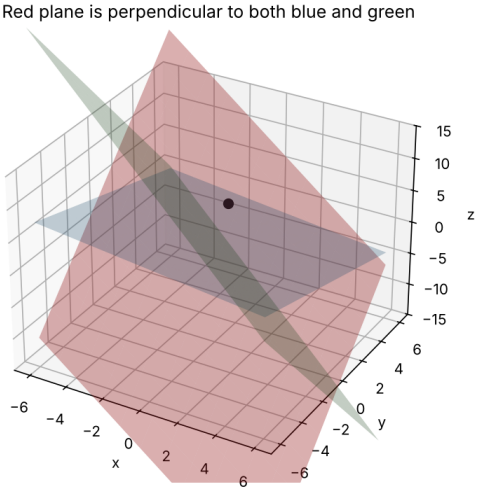


Fig. 2.2.13.1: The required plane (red) with the two given planes

2.2.14 Evaluate: $\int_1^5 \left(|x-1| + |x-2| + |x-4|\right) dx$

Solution: Write each term with the unit step, $|x-a| = (a-x) + 2(x-a)u(x-a)$,

from (2.1.22.2). For $x \geq 0$,

$$f(x) = 7 - 3x + 2\left[(x-1)u(x-1) + (x-2)u(x-2) + (x-4)u(x-4)\right] \tag{2.2.14.1}$$

Its transform from (2.1.21.1) and (2.1.22.2) is

$$F(s) = \frac{7}{s} - \frac{3}{s^2} + \frac{2}{s^2}\left(e^{-s} + e^{-2s} + e^{-4s}\right) \tag{2.2.14.2}$$

Dividing by s as in (2.1.21.2) gives $G(s) = \frac{F(s)}{s}$, and inverting, the integral $g(x) = \int_0^x f$ is

$$g(x) = 7x - \frac{3x^2}{2} + (x-1)^2u(x-1) + (x-2)^2u(x-2) + (x-4)^2u(x-4) \tag{2.2.14.3}$$

$$\int_1^5 f(x)\,dx = g(5) - g(1) = \left(35 - \frac{75}{2} + 16 + 9 + 1\right) - \left(7 - \frac{3}{2}\right) = \frac{47}{2} - \frac{11}{2} = 18 \tag{2.2.14.4}$$

The following code verifies the Laplace result and plots Fig. 2.2.14.1.

codes/2.2.14.py

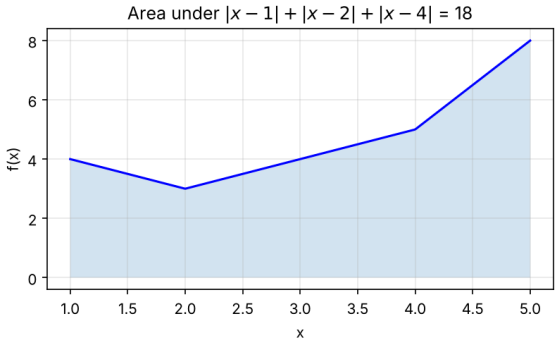


Fig. 2.2.14.1: Area under $|x-1| + |x-2| + |x-4|$ from $x = 1$ to $x = 5$

2.2.15 If x, y, z are different and $\Delta = \begin{vmatrix} x & x^2 & x^3 - 1 \\ y & y^2 & y^3 - 1 \\ z & z^2 & z^3 - 1 \end{vmatrix} = 0$, then using properties of determinants, show that $xyz = 1$.

Solution: Splitting the third column:

$$\Delta = \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} - \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} \tag{2.2.15.1}$$

Taking x, y, z common from the rows of the first determinant, and moving the last column of the second determinant to the front (two column interchanges, so the sign is unchanged):

$$\Delta = xyz\begin{vmatrix}1 & x & x^2\\ 1 & y & y^2\\ 1 & z & z^2\end{vmatrix} - \begin{vmatrix}1 & x & x^2\\ 1 & y & y^2\\ 1 & z & z^2\end{vmatrix} \tag{2.2.15.2}$$

$$= (xyz-1)\begin{vmatrix}1 & x & x^2\\ 1 & y & y^2\\ 1 & z & z^2\end{vmatrix} \tag{2.2.15.3}$$

$$= (xyz-1)(y-x)(z-x)(z-y) = 0 \tag{2.2.15.4}$$

Since x, y, z are different, $(y-x)(z-x)(z-y) \neq 0$. Hence $xyz - 1 = 0$, i.e. $xyz = 1$. The following code factorises Δ symbolically.

codes/2.2.15.py

2.2.16 Prove that: $\sin^{-1}\frac{4}{5} + \tan^{-1}\frac{5}{12} + \cos^{-1}\frac{63}{65} = \frac{\pi}{2}$

Solution: Write $\sin^{-1}\frac{4}{5}$ and $\cos^{-1}\frac{63}{65}$ in terms of $\tan^{-1}$ (from the right triangles with sides $3, 4, 5$ and $16, 63, 65$):

$$\sin^{-1}\frac{4}{5} = \tan^{-1}\frac{4}{3}, \quad \cos^{-1}\frac{63}{65} = \tan^{-1}\frac{16}{63} \tag{2.2.16.1}$$

From (2.1.2.2) (here $ab = \frac{4}{3}\cdot\frac{5}{12} = \frac{5}{9} < 1$):

$$\tan^{-1}\frac{4}{3} + \tan^{-1}\frac{5}{12} = \tan^{-1}\frac{\frac{4}{3}+\frac{5}{12}}{1-\frac{20}{36}} = \tan^{-1}\frac{63}{16} \tag{2.2.16.2}$$

Since $\tan^{-1}\frac{16}{63} = \cot^{-1}\frac{63}{16}$, from (2.1.2.3):

$$\text{LHS} = \tan^{-1}\frac{63}{16} + \cot^{-1}\frac{63}{16} = \frac{\pi}{2} = \text{RHS} \tag{2.2.16.3}$$

Hence proved. The following code verifies each step numerically.

codes/2.2.16.py

2.2.17 Using vectors, find the value of x such that the four points $A(x, 5, -1)$, $B(3, 2, 1)$, $C(4, 5, 5)$ and $D(4, 2, -2)$ are coplanar.

Solution: From (2.1.8.2), the points are coplanar if $\det\begin{bmatrix}\mathbf{AB} & \mathbf{AC} & \mathbf{AD}\end{bmatrix} = 0$. Here

$$\mathbf{AB} = \begin{pmatrix}3-x\\ -3\\ 2\end{pmatrix}, \quad \mathbf{AC} = \begin{pmatrix}4-x\\ 0\\ 6\end{pmatrix}, \quad \mathbf{AD} = \begin{pmatrix}4-x\\ -3\\ -1\end{pmatrix} \tag{2.2.17.1}$$

$$\begin{vmatrix} 3-x & 4-x & 4-x \\ -3 & 0 & -3 \\ 2 & 6 & -1 \end{vmatrix} = 0 \tag{2.2.17.2}$$

$$(3-x)\big[0+18\big] - (4-x)\big[3+6\big] + (4-x)\big[-18-0\big] = 0 \tag{2.2.17.3}$$

$$18(3-x) - 9(4-x) - 18(4-x) = 0 \tag{2.2.17.4}$$

$$54 - 18x - 27(4-x) = 0 \tag{2.2.17.5}$$

$$9x - 54 = 0 \implies x = 6 \tag{2.2.17.6}$$

The following code solves for x, checks that the determinant vanishes at $x = 6$ and plots Fig. 2.2.17.1.

codes/2.2.17.py

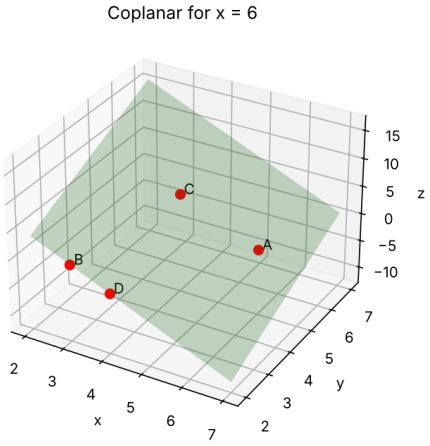


Fig. 2.2.17.1: The four coplanar points for $x = 6$

2.2.18 (a) Differentiate $\tan^{-1}\dfrac{3x-x^3}{1-3x^2}$, $|x| < \dfrac{1}{\sqrt{3}}$ w.r.t. $\tan^{-1}\dfrac{x}{\sqrt{1-x^2}}$.

OR

(b) If $\sqrt{1-x^2}+\sqrt{1-y^2} = a(x-y)$, $|x|<1$, $|y|<1$, show that $\dfrac{dy}{dx} = \sqrt{\dfrac{1-y^2}{1-x^2}}$.

(a) Solution: Let $u = \tan^{-1}\frac{3x-x^3}{1-3x^2}$ and $v = \tan^{-1}\frac{x}{\sqrt{1-x^2}}$. Put $x = \tan\phi$; since $|x| < \frac{1}{\sqrt{3}}$, $|3\phi| < \frac{\pi}{2}$ and

$$u = \tan^{-1}(\tan 3\phi) = 3\phi = 3\tan^{-1}x \implies \frac{du}{dx} = \frac{3}{1+x^2} \tag{2.2.18.1}$$

From (2.1.2.4), $v = \sin^{-1} x$, so $\frac{dv}{dx} = \frac{1}{\sqrt{1-x^2}}$. Hence

$$\frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{3\sqrt{1-x^2}}{1+x^2} \tag{2.2.18.2}$$

(b) Solution: Put $x = \sin\alpha$, $y = \sin\beta$ with $\alpha, \beta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$:

$$\cos\alpha + \cos\beta = a(\sin\alpha - \sin\beta) \tag{2.2.18.3}$$

$$2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2} = 2a\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} \tag{2.2.18.4}$$

$$\implies \cot\frac{\alpha-\beta}{2} = a \implies \sin^{-1} x - \sin^{-1} y = 2\cot^{-1} a \tag{2.2.18.5}$$

Differentiating w.r.t. x (the right side is a constant):

$$\frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-y^2}}\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}} \tag{2.2.18.6}$$

The following code verifies both parts and plots Fig. 2.2.18.1.

codes/2.2.18.py

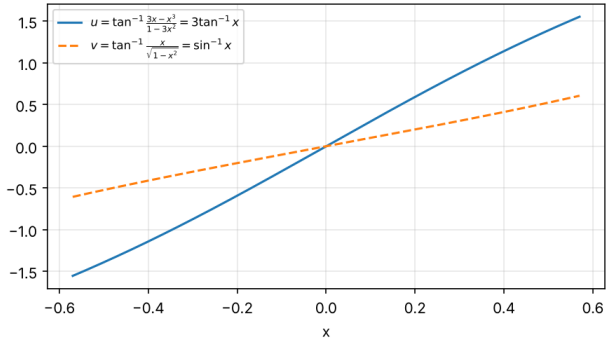


Fig. 2.2.18.1: The functions $u = 3\tan^{-1} x$ and $v = \sin^{-1} x$

2.2.19 (a) Prove that the relation R in the set $A = \{1, 2, 3, 4, 5, 6, 7\}$ given by $R = \{(a, b) : |a - b| \text{ is even}\}$ is an equivalence relation.

OR

(b) Show that the function f in $A = \mathbb{R} - \left\{\frac{2}{3}\right\}$ defined as $f(x) = \frac{4x+3}{6x-4}$ is one-one and onto. Hence, find f^{-1}.

(a) Solution: By 2.1.12.1 we check three properties.

$$\text{Reflexive: } |a-a| = 0 \text{ is even } \implies (a,a) \in R \tag{2.2.19.1}$$

$$\text{Symmetric: } |a-b| = |b-a| \implies (a,b) \in R \Rightarrow (b,a) \in R \tag{2.2.19.2}$$

$$\text{Transitive: } |a-b|, |b-c| \text{ even } \implies a-c = (a-b)+(b-c) \text{ is even} \tag{2.2.19.3}$$

Hence R is an equivalence relation.

(b) Solution: *One-one:* Let $f(x_1) = f(x_2)$:

$$(4x_1+3)(6x_2-4) = (4x_2+3)(6x_1-4) \tag{2.2.19.4}$$

$$24x_1x_2 - 16x_1 + 18x_2 - 12 = 24x_1x_2 - 16x_2 + 18x_1 - 12 \tag{2.2.19.5}$$

$$34x_2 = 34x_1 \implies x_1 = x_2 \tag{2.2.19.6}$$

Onto: Let $y = \frac{4x+3}{6x-4}$, $y \neq \frac{2}{3}$. Solving for x:

$$6xy - 4y = 4x + 3 \implies x = \frac{4y+3}{6y-4} \tag{2.2.19.7}$$

This x is defined for every $y \neq \frac{2}{3}$, and $x \neq \frac{2}{3}$ (otherwise $12y+9 = 12y-8$, which is impossible). So every $y \in A$ has a pre-image and f is onto. Hence

$$f^{-1}(y) = \frac{4y+3}{6y-4}, \quad \text{i.e. } f^{-1} = f \tag{2.2.19.8}$$

The following code verifies both parts and generates Fig. 2.2.19.1.

codes/2.2.19.py

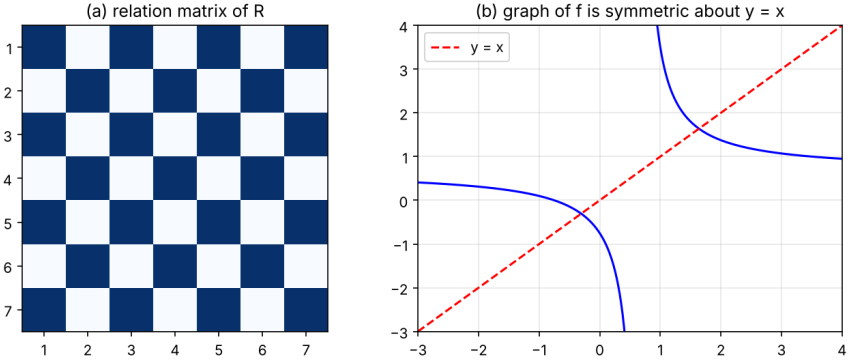


Fig. 2.2.19.1: (a) Relation matrix of R; (b) graph of f, symmetric about $y = x$

2.2.20 (a) Find the particular solution of the differential equation $(1+e^{2x})\,dy + (1+y^2)e^x\,dx = 0$, given that $y(0) = 1$.

OR

(b) Find the particular solution of the differential equation $x\dfrac{dy}{dx}\sin\left(\dfrac{y}{x}\right)+x-y\sin\left(\dfrac{y}{x}\right)=0$, given that $y(1)=\dfrac{\pi}{2}$.

(a) Solution: Separating the variables:

$$\frac{dy}{1+y^2}=-\frac{e^x\,dx}{1+e^{2x}} \tag{2.2.20.1}$$

$$\tan^{-1}y=-\tan^{-1}(e^x)+C \tag{2.2.20.2}$$

Using $y(0)=1$: $\frac{\pi}{4}=-\frac{\pi}{4}+C\implies C=\frac{\pi}{2}$. Then from (2.1.2.3),

$$\tan^{-1}y=\frac{\pi}{2}-\tan^{-1}(e^x)=\cot^{-1}(e^x)=\tan^{-1}(e^{-x})\implies y=e^{-x} \tag{2.2.20.3}$$

(b) Solution: Dividing by $x\sin\left(\frac{y}{x}\right)$:

$$\frac{dy}{dx}=\frac{y}{x}-\frac{1}{\sin(y/x)} \tag{2.2.20.4}$$

This is homogeneous. From (2.1.6.1), put $y=vx$:

$$v+x\frac{dv}{dx}=v-\frac{1}{\sin v} \tag{2.2.20.5}$$

$$\sin v\,dv=-\frac{dx}{x} \tag{2.2.20.6}$$

$$-\cos v=-\ln|x|+C\implies\cos\left(\frac{y}{x}\right)=\ln|x|+C' \tag{2.2.20.7}$$

Using $y(1)=\frac{\pi}{2}$: $\cos\frac{\pi}{2}=0=\ln 1+C'\implies C'=0$. Hence

$$\cos\left(\frac{y}{x}\right)=\ln|x| \tag{2.2.20.8}$$

The following code verifies both solutions and plots Fig. 2.2.20.1.

codes/2.2.20.py

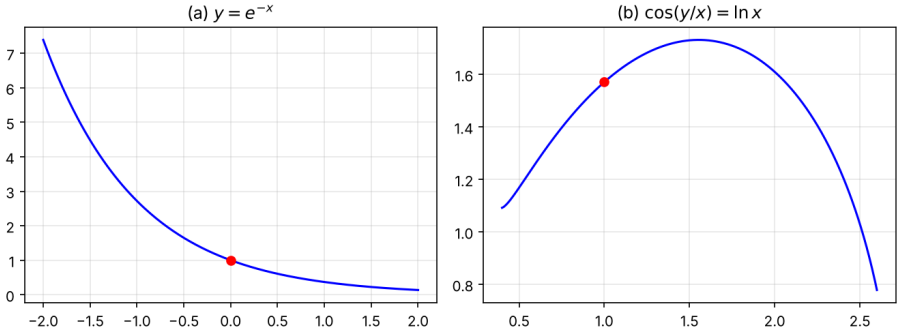


Fig. 2.2.20.1: (a) $y=e^{-x}$ and (b) $\cos(y/x)=\ln x$

2.2.21 If $y = (\sin x)^x + \sin^{-1}\left(\sqrt{1-x^2}\right)$, then find $\dfrac{dy}{dx}$.

Solution: The function is defined for $0 < x < 1$ (we need $\sin x > 0$ and $|x| \leq 1$). Let $y = u + w$ with $u = (\sin x)^x$ and $w = \sin^{-1}\sqrt{1-x^2}$. From (2.1.13.1):

$$\ln u = x \ln \sin x \implies \frac{1}{u}\frac{du}{dx} = \ln \sin x + x \cot x \tag{2.2.21.1}$$

$$\implies \frac{du}{dx} = (\sin x)^x \left(x \cot x + \ln \sin x\right) \tag{2.2.21.2}$$

For $0 < x < 1$ put $x = \cos\phi$, $\phi \in \left(0, \frac{\pi}{2}\right)$; then $\sqrt{1-x^2} = \sin\phi$ and $w = \phi = \cos^{-1} x$:

$$\frac{dw}{dx} = -\frac{1}{\sqrt{1-x^2}} \tag{2.2.21.3}$$

Hence

$$\frac{dy}{dx} = (\sin x)^x \left(x \cot x + \ln \sin x\right) - \frac{1}{\sqrt{1-x^2}} \tag{2.2.21.4}$$

The following code verifies the derivative and plots Fig. 2.2.21.1.

codes/2.2.21.py

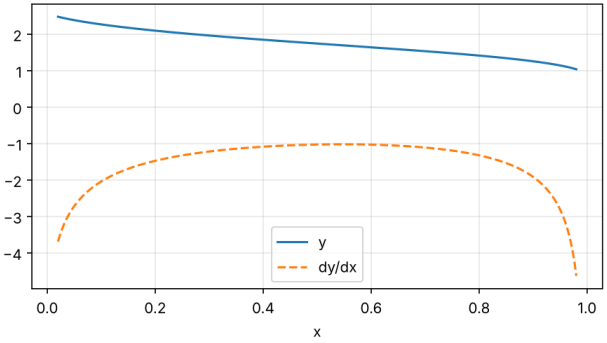


Fig. 2.2.21.1: The function y and its derivative on $(0, 1)$

2.2.22 Find: $\displaystyle\int \cos 2x \cos 4x \cos 6x \, dx$

Solution: From (2.1.9.4), the integrand is a sum of cosines:

$$f(x) = \cos 2x \cos 4x \cos 6x = \frac{1}{4}\left(1 + \cos 12x + \cos 8x + \cos 4x\right) \tag{2.2.22.1}$$

Taking the Laplace transform with (2.1.21.1),

$$F(s) = \frac{1}{4}\left[\frac{1}{s} + \frac{s}{s^2+144} + \frac{s}{s^2+64} + \frac{s}{s^2+16}\right] \tag{2.2.22.2}$$

Dividing by s as in (2.1.21.2) gives the transform of the integral:

$$G(s) = \frac{F(s)}{s} = \frac{1}{4}\left[\frac{1}{s^2} + \frac{1}{s^2+144} + \frac{1}{s^2+64} + \frac{1}{s^2+16}\right] \tag{2.2.22.3}$$

Using $\mathcal{L}^{-1}\left\{\frac{1}{s^2+\omega^2}\right\} = \frac{\sin \omega x}{\omega}$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = x$,

$$\int \cos 2x \cos 4x \cos 6x \, dx = \frac{1}{4}\left[x + \frac{\sin 12x}{12} + \frac{\sin 8x}{8} + \frac{\sin 4x}{4}\right] + C \tag{2.2.22.4}$$

The following code computes the Laplace transform and its inverse, and checks the result by differentiation.

codes/2.2.22.py

2.2.23 Find the interval in which the function f given by $f(x) = \sin 2x + \cos 2x$, $0 \le x \le \pi$ is strictly decreasing.

Solution: Differentiating,

$$f'(x) = 2\cos 2x - 2\sin 2x = 2\sqrt{2}\cos\left(2x + \frac{\pi}{4}\right) \tag{2.2.23.1}$$

From 2.1.14.1, f is strictly decreasing where $f'(x) < 0$. For $0 \le x \le \pi$, $2x + \frac{\pi}{4} \in \left[\frac{\pi}{4}, \frac{9\pi}{4}\right]$, and $\cos\theta < 0$ there only for $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$:

$$\frac{\pi}{2} < 2x + \frac{\pi}{4} < \frac{3\pi}{2} \implies \frac{\pi}{8} < x < \frac{5\pi}{8} \tag{2.2.23.2}$$

Thus f is strictly decreasing on $\left(\frac{\pi}{8}, \frac{5\pi}{8}\right)$. The following code verifies the interval and plots Fig. 2.2.23.1.

codes/2.2.23.py

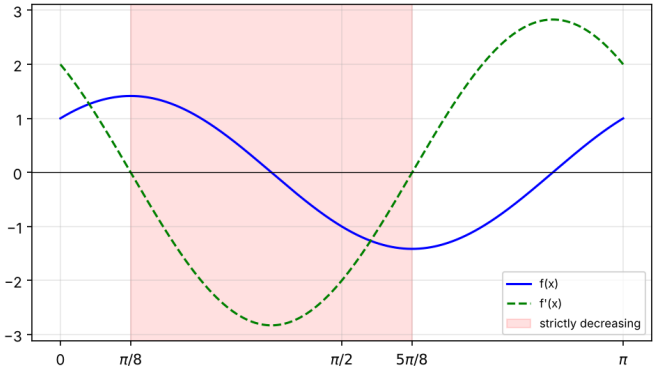


Fig. 2.2.23.1: $f(x) = \sin 2x + \cos 2x$ and $f'(x)$

2.2.24 (a) Using elementary row transformations, find the inverse of the matrix

$$\begin{bmatrix} 3 & 0 & -1 \\ 2 & 3 & 0 \\ 0 & 4 & 1 \end{bmatrix}$$

OR

(b) Using matrices, solve the following system of linear equations:

$$\begin{aligned} 2x + 3y + 10z &= 4 \\ 4x - 6y + 5z &= 1 \\ 6x + 9y - 20z &= 2 \end{aligned}$$

(a) Solution: From (2.1.11.2), apply row operations to $[A \,|\, I]$:

$$[A|I] = \left[\begin{array}{ccc|ccc} 3 & 0 & -1 & 1 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 4 & 1 & 0 & 0 & 1 \end{array}\right] \tag{2.2.24.1}$$

$$\xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & -3 & -1 & 1 & -1 & 0 \\ 2 & 3 & 0 & 0 & 1 & 0 \\ 0 & 4 & 1 & 0 & 0 & 1 \end{array}\right] \tag{2.2.24.2}$$

$$\xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & -3 & -1 & 1 & -1 & 0 \\ 0 & 9 & 2 & -2 & 3 & 0 \\ 0 & 4 & 1 & 0 & 0 & 1 \end{array}\right] \tag{2.2.24.3}$$

$$\xrightarrow{R_2 \to R_2 - 2R_3} \left[\begin{array}{ccc|ccc} 1 & -3 & -1 & 1 & -1 & 0 \\ 0 & 1 & 0 & -2 & 3 & -2 \\ 0 & 4 & 1 & 0 & 0 & 1 \end{array}\right] \tag{2.2.24.4}$$

$$\xrightarrow{R_1 \to R_1 + 3R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & -5 & 8 & -6 \\ 0 & 1 & 0 & -2 & 3 & -2 \\ 0 & 4 & 1 & 0 & 0 & 1 \end{array}\right] \tag{2.2.24.5}$$

$$\xrightarrow{R_3 \to R_3 - 4R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & -5 & 8 & -6 \\ 0 & 1 & 0 & -2 & 3 & -2 \\ 0 & 0 & 1 & 8 & -12 & 9 \end{array}\right] \tag{2.2.24.6}$$

$$\xrightarrow{R_1 \to R_1 + R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -4 & 3 \\ 0 & 1 & 0 & -2 & 3 & -2 \\ 0 & 0 & 1 & 8 & -12 & 9 \end{array}\right] \tag{2.2.24.7}$$

Thus,

$$A^{-1} = \begin{bmatrix} 3 & -4 & 3 \\ -2 & 3 & -2 \\ 8 & -12 & 9 \end{bmatrix} \tag{2.2.24.8}$$

(b) Solution: Write the system as $A\mathbf{x} = \mathbf{b}$ and use Gaussian elimination on $[A \,|\, \mathbf{b}]$,

with the multipliers of (2.1.19.1):

$$\left[A \,|\, \mathbf{b}\right] = \left[\begin{array}{ccc|c} 2 & 3 & 10 & 4 \\ 4 & -6 & 5 & 1 \\ 6 & 9 & -20 & 2 \end{array}\right] \tag{2.2.24.9}$$

$$\xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 2 & 3 & 10 & 4 \\ 0 & -12 & -15 & -7 \\ 0 & 0 & -50 & -10 \end{array}\right] \tag{2.2.24.10}$$

The pivots 2, -12 and -50 are all nonzero, so the elimination does not break down and the solution is unique. By back-substitution, from (2.1.19.2),

$$-50z = -10 \implies z = \frac{1}{5} \tag{2.2.24.11}$$

$$-12y - 15\left(\frac{1}{5}\right) = -7 \implies y = \frac{1}{3} \tag{2.2.24.12}$$

$$2x + 3\left(\frac{1}{3}\right) + 10\left(\frac{1}{5}\right) = 4 \implies x = \frac{1}{2} \tag{2.2.24.13}$$

(2.1.19.1, 2.1.19.2)

Thus $x = \frac{1}{2}$, $y = \frac{1}{3}$, $z = \frac{1}{5}$. The C program carries out the elimination and back-substitution and checks the residual of the original system:

codes/2.2.24b.c

To see how the cost grows with the size n of the system, a second C program times Gaussian elimination on random $n \times n$ systems and writes the times to a file. Python is used only to plot them in Fig. 2.2.24.1; the time grows as n^3, as in (2.1.19.3).

codes/2.2.24b_time.c

codes/2.2.24b_time.py

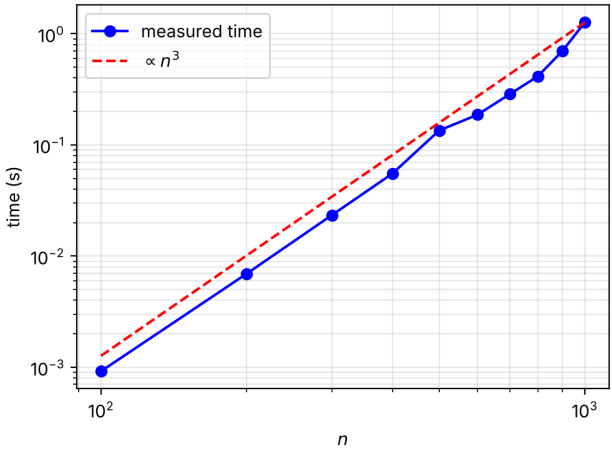


Fig. 2.2.24.1: Time of Gaussian elimination against n (log-log), measured by the C program

The Python program below verifies the inverse in (a) and the solution in (b).

codes/2.2.24a.py

2.2.25 (a) Using integration, find the area of the following region: $\{(x, y) : x^2 + y^2 \leq 16a^2 \text{ and } y^2 \leq 6ax\}$

OR

(b) Using integration, find the area of triangle ABC bounded by the lines $4x-y+5 = 0$, $x + y - 5 = 0$ and $x - 4y + 5 = 0$.

(a) Solution: The circle $x^2 + y^2 = 16a^2$ and the parabola $y^2 = 6ax$ meet where

$$x^2 + 6ax - 16a^2 = 0 \implies (x + 8a)(x - 2a) = 0 \implies x = 2a, \quad y = \pm 2\sqrt{3}\, a \tag{2.2.25.1}$$

Writing both curves as x in terms of y, the parabola $x = \frac{y^2}{6a}$ is the left boundary and the circle $x = \sqrt{16a^2 - y^2}$ is the right boundary. The region is symmetric about the x-axis, so from (2.1.15.2),

$$\text{Area} = 2\int_0^{2\sqrt{3}a} \left(\sqrt{16a^2 - y^2} - \frac{y^2}{6a}\right) dy \tag{2.2.25.2}$$

From (2.1.9.3),

$$= 2\left[\frac{y}{2}\sqrt{16a^2 - y^2} + 8a^2 \sin^{-1}\frac{y}{4a} - \frac{y^3}{18a}\right]_0^{2\sqrt{3}a} \tag{2.2.25.3}$$

$$= 2\left[2\sqrt{3}a^2 + \frac{8\pi a^2}{3} - \frac{4\sqrt{3}a^2}{3}\right] \tag{2.2.25.4}$$

$$= \frac{4a^2}{3}\left(4\pi + \sqrt{3}\right) \text{ sq. units} \tag{2.2.25.5}$$

(b) Solution: Solving the lines pairwise gives the vertices $A(-1,1)$ (lines 1 and 3), $B(0,5)$ (lines 1 and 2) and $C(3,2)$ (lines 2 and 3). Put $t = x+1$. The height of the triangle at t is the upper boundary minus the lower boundary $y = \frac{x+5}{4}$: it is $\frac{15}{4}t$ for $0 \le t \le 1$, and $5 - \frac{5}{4}t$ for $1 \le t \le 4$. With the unit step,

$$h(t) = \frac{15}{4}t - 5(t-1)u(t-1) + \frac{5}{4}(t-4)u(t-4) \tag{2.2.25.6}$$

From (2.1.21.1) and (2.1.22.2), and dividing by s as in (2.1.21.2),

$$H(s) = \frac{1}{s^2}\left(\frac{15}{4} - 5e^{-s} + \frac{5}{4}e^{-4s}\right) \implies \frac{H(s)}{s} = \frac{1}{s^3}\left(\frac{15}{4} - 5e^{-s} + \frac{5}{4}e^{-4s}\right) \tag{2.2.25.7}$$

Inverting, the area up to t is $\frac{15}{8}t^2 - \frac{5}{2}(t-1)^2u(t-1) + \frac{5}{8}(t-4)^2u(t-4)$, so

$$\text{Area} = \int_0^4 h(t)\,dt = \frac{15}{8}(16) - \frac{5}{2}(9) = 30 - \frac{45}{2} = \frac{15}{2} \text{ sq. units} \tag{2.2.25.8}$$

This agrees with (2.1.15.3). The following code verifies both parts and generates Fig. 2.2.25.1.

codes/2.2.25.py

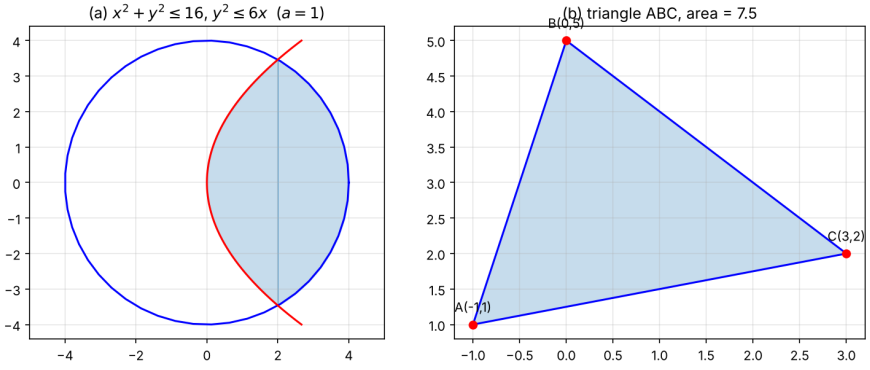


Fig. 2.2.25.1: (a) Region inside the circle and the parabola ($a = 1$); (b) triangle ABC

2.2.26 A company manufactures two types of novelty souvenirs made of plywood. Souvenirs of type A require 5 minutes each for cutting and 10 minutes each for assembling. Souvenirs of type B require 8 minutes each for cutting and 8 minutes each for assembling. There are 3 hours 20 minutes available for cutting and 4 hours for assembling. The profit for type A souvenirs is ₹100 each and for type B souvenirs, profit is ₹120 each. How many souvenirs of each type should the company manufacture in order to maximise the profit? Formulate the problem as a LPP and then solve it graphically.

Solution: Let the company make x souvenirs of type A and y of type B. The available times are 3 h 20 min $= 200$ minutes for cutting and 4 h $= 240$ minutes for assembling. From (2.1.17.1):

$$\max Z = 100x + 120y \quad \text{s.t.} \quad 5x + 8y \le 200 \tag{2.2.26.1}$$

$$10x + 8y \le 240 \tag{2.2.26.2}$$

$$x,\ y \ge 0 \tag{2.2.26.3}$$

The corner points of the feasible region are $O(0,0)$, $(24,0)$, $(8,20)$ (where $5x+8y = 200$ meets $10x + 8y = 240$) and $(0,25)$:

Corner point	$Z = 100x + 120y$
$(0,0)$	0
$(24,0)$	2400
$(8,20)$	3200
$(0,25)$	3000

The maximum profit is ₹3200, obtained by making 8 souvenirs of type A and 20 souvenirs of type B. The following code finds the corner points, checks the result with a linear-programming solver and plots Fig. 2.2.26.1.

codes/2.2.26.py

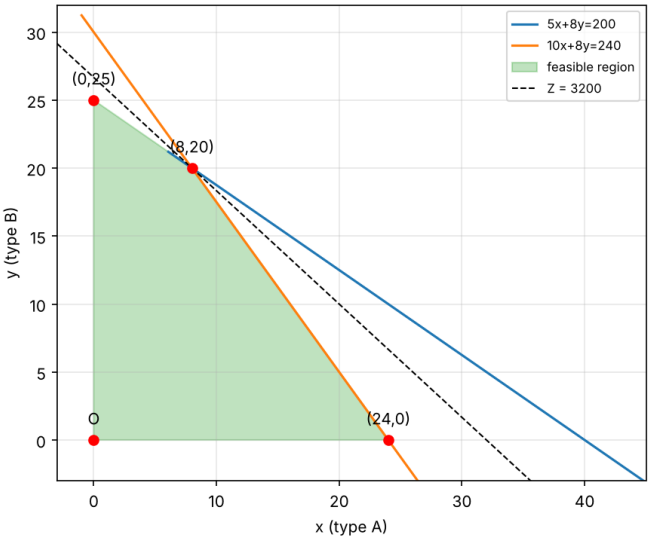


Fig. 2.2.26.1: Feasible region and the optimum at $(8, 20)$

2.2.27 (a) Find the vector equation of the line passing through $(2, 1, -1)$ and parallel to the line $\mathbf{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$. Also, find the distance between these two lines.

OR

(b) Find the coordinates of the foot Q of the perpendicular drawn from the point $P(1, 3, 4)$ to the plane $2x - y + z + 3 = 0$. Find the distance PQ and the image of P treating the plane as a mirror.

(a) Solution: The required line has the same direction $\mathbf{b} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$ and passes through $\mathbf{a}_2 = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$:

$$\mathbf{r} = (2\hat{i} + \hat{j} - \hat{k}) + \mu(2\hat{i} - \hat{j} + \hat{k}) \tag{2.2.27.1}$$

With $\mathbf{a}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{a}_2 - \mathbf{a}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and

$$(\mathbf{a}_2 - \mathbf{a}_1) \times \mathbf{b} = \begin{pmatrix} 0 \cdot 1 - (-1)(-1) \\ (-1)(2) - 1 \cdot 1 \\ 1 \cdot (-1) - 0 \cdot 2 \end{pmatrix} = \begin{pmatrix} -1 \\ -3 \\ -1 \end{pmatrix} \tag{2.2.27.2}$$

From (2.1.16.1):

$$d = \frac{\sqrt{1+9+1}}{\sqrt{4+1+1}} = \sqrt{\frac{11}{6}} \approx 1.354 \text{ units} \tag{2.2.27.3}$$

(b) Solution: The normal of the plane is $\mathbf{n} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$ and $c = -3$. From (2.1.16.2):

$$t = \frac{-3-(2-3+4)}{4+1+1} = \frac{-6}{6} = -1 \tag{2.2.27.4}$$

$$\mathbf{Q} = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix} \tag{2.2.27.5}$$

$$PQ = |t| \, \|\mathbf{n}\| = \sqrt{6} \tag{2.2.27.6}$$

$$\mathbf{P}' = \mathbf{P} + 2t\mathbf{n} = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} - 2\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 5 \\ 2 \end{pmatrix} \tag{2.2.27.7}$$

Thus $Q(-1,4,3)$, $PQ = \sqrt{6}$ and the image of P is $P'(-3,5,2)$. The following code verifies both parts and generates Fig. 2.2.27.1.

codes/2.2.27.py

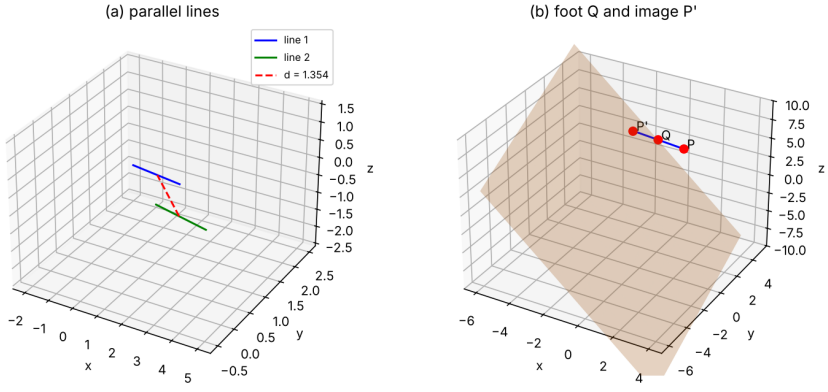


Fig. 2.2.27.1: (a) The two parallel lines; (b) foot Q and image P' of P

2.2.28 A card from a pack of 52 playing cards is lost. From the remaining cards of the pack, two cards are drawn at random (without replacement) and both are found to be spades. Find the probability of the lost card being a spade.

Solution: Let $L = 1$ if the lost card is a spade and $L = 0$ otherwise, and let Y be the number of spades among the two cards drawn. The joint PMF is $p_{LY}(l,2) =$

$p_L(l)\,p_{Y|L}(2|l)$, with

$$p_L(1)=\frac{1}{4},\quad p_L(0)=\frac{3}{4} \tag{2.2.28.1}$$

If a spade is lost, 12 spades remain among 51 cards; otherwise 13 remain:

$$p_{LY}(1,2)=\frac{1}{4}\cdot\frac{12}{51}\cdot\frac{11}{50},\quad p_{LY}(0,2)=\frac{3}{4}\cdot\frac{13}{51}\cdot\frac{12}{50} \tag{2.2.28.2}$$

Normalizing over l using (2.1.20.1), $p_{L|Y}(1|2)=\dfrac{p_{LY}(1,2)}{p_{LY}(1,2)+p_{LY}(0,2)}$:

$$\Pr\left(L=1\mid Y=2\right)=\frac{132}{132+468}=\frac{11}{50} \tag{2.2.28.3}$$

The following code verifies the result with exact fractions and a simulation.

codes/2.2.28.py

2.2.29 Prove that the radius of the right circular cylinder of greatest curved surface area which can be inscribed in a given cone is half of that of the cone.

Solution: Let the cone have base radius R and height H, and let the inscribed cylinder have radius r and height h. By similar triangles in the cross-section, from (2.1.18.1),

$$\frac{h}{H}=\frac{R-r}{R}\implies h=\frac{H(R-r)}{R} \tag{2.2.29.1}$$

The curved surface area is

$$S=2\pi rh=\frac{2\pi H}{R}\left(Rr-r^2\right) \tag{2.2.29.2}$$

Differentiating,

$$\frac{dS}{dr}=\frac{2\pi H}{R}(R-2r)=0\implies r=\frac{R}{2} \tag{2.2.29.3}$$

$$\frac{d^2S}{dr^2}=-\frac{4\pi H}{R}<0 \tag{2.2.29.4}$$

So S is maximum at $r=\frac{R}{2}$, i.e. the radius of the cylinder is half the radius of the cone (and its height is $\frac{H}{2}$). Hence proved. The following code verifies the maximum and plots Fig. 2.2.29.1.

codes/2.2.29.py

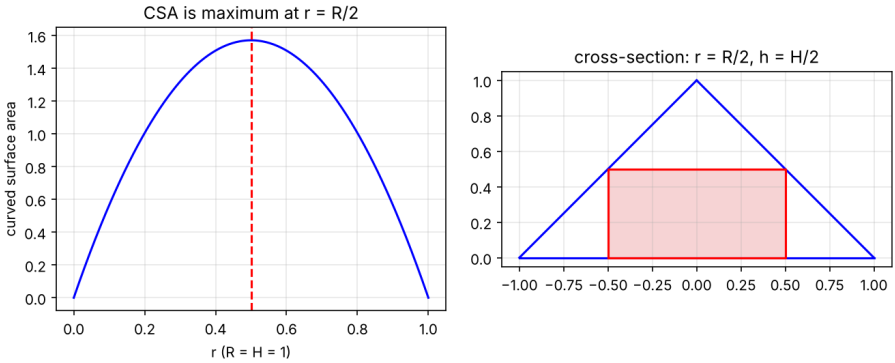


Fig. 2.2.29.1: (a) Curved surface area against r; (b) the cylinder with $r = R/2$ inside the cone